Improving Student Outcomes through Adaptive Learning Platforms: Experimental Evidence from the Dominican Republic

Carolina Lopez
Development Research Group, World Bank

Astrid Pineda
World Bank *

May 31, 2026

Click here for the latest version

Abstract

Most students in developing countries receive grade-level instruction despite lacking prerequisite skills, widening learning gaps. To address this mismatch, we conducted a randomized controlled trial integrating computer-adaptive learning (CAL) technology into regular classroom instruction with 1,333 ninth-grade students in the Dominican Republic. We replaced two of seven weekly mathematics hours with an adaptive software that adapts to individual skill levels. CAL improved test scores by 0.28-0.31 standard deviations. However, we find no evidence that adding small-group tutoring enhanced these benefits; if anything, classrooms assigned to receive both interventions showed smaller gains than those receiving CAL alone, though this difference is not statistically significant. We document implementation challenges and explore potential mechanisms, including reduced platform engagement in tutoring classrooms. These results suggest computer-adaptive learning software can be effectively integrated into classroom instruction to improve learning, but combining multiple interventions may produce unintended behavioral responses that diminish their effectiveness.

*We thank the Ministry of Education of the Dominican Republic, especially Ancell Scheker Mendoza, Vice Minister of Technical and Pedagogical Affairs, and the participating schools for facilitating this research. We are specially grateful to Sarah Cohodes, Alex Eble, Alejandro Ganimian, Juan Barón, Matteo Magnaricotte, Peter Bergman, Donald Green and Chris Neilson for valuable comments and support. Aurora Salvador provided excellent research assistance. The study was funded by the Tinker Foundation and is registered as AEARCTR-0012383 in the AER registry. It was approved under protocol number 24-444 by the Teachers College IRB. Pineda was supported by the Institute of Education Sciences, U.S. Department of Education, through grant R305B200017 to Teachers College Columbia University. The findings, interpretations, and conclusions expressed in this paper are entirely those of the authors. They do not necessarily represent the views of the World Bank and its affiliated organizations, or those of the Executive Directors of the World Bank or the governments they represent. All errors are our own. Lopez: World Bank, carolina_lopez@worldbank.org, Pineda: World Bank, apineda@worldbank.org

1 Introduction

Most students in developing countries receive grade-level instruction despite lacking prerequisite skills, and this mismatch widens learning gaps over time (Muralidharan et al., 2019). The problem is widespread: less than 20 percent of students in low- and middle-income countries meet minimum proficiency standards in mathematics (World Bank, 2017), and the typical student in Latin America performs approximately five years behind their OECD counterparts (IDB, 2023). When students who lack foundational knowledge are taught content beyond their current level, they fall further behind rather than catch up. This dynamic contributes to both poor achievement and high dropout rates.

Computer-adaptive learning (CAL) platforms offer a potentially scalable solution to this instructional mismatch. By continuously assessing student knowledge and adjusting content difficulty accordingly, CAL can deliver personalized instruction at a fraction of the cost of one-on-one tutoring. Recent evidence from India demonstrates that CAL can generate substantial learning gains when substituted for regular classroom instruction, with effects ranging from 0.20–0.22 standard deviations at scale (Muralidharan and Singh, 2025) to 0.45 standard deviations under intensive implementation support (Oreopoulos et al., 2026). Yet as policymakers seek to maximize learning recovery, a critical question remains unexplored: can combining CAL with other effective interventions—such as small-group tutoring—generate complementary benefits, or might such combinations produce unintended interference?

This paper addresses this question through a randomized controlled trial integrating CAL into regular mathematics instruction for 1,333 ninth-grade students across 38 classrooms in the Dominican Republic. We replace two of seven weekly math hours with ALEKS, a learning platform, testing whether adaptive technology can improve learning when substituted for traditional instruction rather than added to it. Our design also enables a direct test of intervention complementarities: within treated classrooms, we randomly assign small-group tutoring to academically at-risk students, enabling us to examine potential complementarities between technology-based and traditional instructional approaches.

In 2023, we implemented our intervention by integrating CAL into regular math instruction for ninth graders, with randomization conducted at the classroom level across 38 classrooms. We employed ALEKS, a CAL platform developed by McGraw Hill that had been adapted to the Dominican Republic's math curriculum and previously deployed in approximately 60 schools nationwide.[1] All students in treated classrooms received licenses to access the software during two of their seven weekly math instruction hours. To explore potential complementarities between adaptive technology and traditional instruction methods, we implemented a second randomization within treated classrooms, providing up to three weekly hours of small-group tutoring to students

[1] The intervention was implemented by the Ministry of Education from 2018 to 2023, with support from the World Bank.

identified as academically at-risk. Tutoring sessions were scheduled during existing extracurricular periods to avoid confounding effects from additional instructional time.

We find that the use of the adaptive software during math classes improved student outcomes by 0.29 to 0.31 standard deviations by endline (six months after the start of the intervention). These positive effects are robust across various model specifications and consistent across multiple knowledge domains, including algebra and geometry, as well as different difficulty levels, such as content below and at grade level. However, we do not find statistically significant differences between classes that received the tutoring component and the control group, suggesting that adding tutoring to the CAL component may have attenuated the effectiveness of the latter.

Why did the two treatment arms have different impacts? Implementation fidelity does not explain the difference: at an aggregate level, both CAL and tutoring faced similar compliance rates (approximately 65–68 percent of intended dosage) and similar barriers, predominantly school closures and infrastructure failures. Instead, the evidence points to behavioral mechanisms. Students in tutoring classrooms—including non-tutees—were less likely to use the adaptive platform, exhibited lower levels of locus of control (i.e., belief in their ability to influence their own outcomes), and experienced lower levels of math anxiety compared to the CAL-only group. We also find evidence indicating that the tutoring may not have been well-targeted, as students selected by teachers to receive tutoring were not among the lowest performers at baseline, as was intended by the intervention.

This paper makes three contributions to the literature on educational technology in developing countries. First, we provide experimental evidence on CAL as a *substitute* for—rather than a supplement to—traditional instruction. Increasing access to computer-adaptive technologies has spurred a growing body of research (Angel-Urdinola et al., 2023; Guryan et al., 2023; Muralidharan et al., 2019).[2] Most existing studies have evaluated CAL in after-school settings or combined with tutoring (Muralidharan et al., 2019; Guryan et al., 2023), leaving open questions about its effectiveness when integrated into regular school schedules or implemented alone (Muralidharan et al., 2019). Recent evidence from India demonstrates that intensive implementation support can generate CAL effects of 0.45 standard deviations when dedicated personnel ensure high-fidelity platform usage (Oreopoulos et al., 2026), but such models require dedicated staff and face scalability constraints. Furthermore, the impact of CAL alone versus additional instructional time remains difficult to disentangle (Glewwe and Muralidharan, 2016; Muralidharan et al., 2019). To our knowledge, this study is among the first to test the substitution of regular math instruction with the use of an adaptive learning platform in a developing country, alongside recent work from India (Muralidharan and Singh, 2023, 2025). By replacing two of seven weekly math instruction hours with CAL sessions, we offer insight into its efficacy when implemented as part of the standard

[2] We focus on computer-adaptive learning, which personalizes content to students' actual learning levels, as opposed to traditional computer-assisted instruction, which reviews only grade-appropriate content. Studies of the latter include Angrist and Lavy (2002), Barrow et al. (2009), and Linden et al. (2003).

school curriculum without increasing total instructional time.

Second, we provide a rare experimental test of intervention complementarities in education. Tutoring has been shown to be one of the most effective interventions for improving learning outcomes. A recent meta-analysis by Nickow et al. (2020) reported an average effect size of 0.37 standard deviations for tutoring interventions, with one-to-one tutoring showing the largest effects. However, the scalability of this approach is hindered by high costs and the challenge of recruiting qualified tutors (Kraft and Falken, 2020; Nickow et al., 2020; Kraft et al., 2024). The longstanding issue of finding scalable alternatives that are as effective—known as the "2-sigma problem" (Bloom, 1984)—remains unresolved, though recent evidence suggests that small-group tutoring can achieve similar results at lower costs (Augustine et al., 2016; Guryan et al., 2023; Kraft and Falken, 2020; Nickow et al., 2020; Kraft et al., 2024). While the literature often assumes that combining effective interventions yields additive benefits, this assumption is rarely tested experimentally. Our two-stage randomization design enables direct estimation of the CAL×Tutoring interaction, providing causal evidence on intervention complementarities.

Third, we document behavioral mechanisms through which intervention complementarities may fail. Contrary to expectations, we find that in-person tutoring did not enhance the impact of the CAL intervention; the combined treatment produced effects statistically indistinguishable from CAL alone and, if anything, smaller in magnitude. We show that students in tutoring classrooms—including those not directly receiving tutoring—exhibited reduced platform engagement, higher levels of math anxiety, and lower locus of control relative to the CAL-only group. These findings suggest that human support may crowd out the self-directed learning processes that make adaptive technology effective, highlighting the importance of empirically testing intervention bundles rather than assuming additive effects.

The deepening of the learning crisis due to COVID-19 highlights the need for more resilient education systems that can better withstand disruptions. Our findings demonstrate that adaptive learning technology can deliver substantial learning gains even amid infrastructure constraints typical of developing country settings. However, the reduced effectiveness we observe when combining CAL with tutoring—linked to decreased platform engagement and shifts in student agency—underscores that simply layering interventions does not guarantee additive effects. Successful implementation requires careful attention to how different components interact, not merely the addition of more resources.

This paper proceeds as follows. In Section 2, we describe in detail the context, the intervention, implementation, and its components. In Section 3, we describe our data. In Section 4 and 5, we present the empirical framework and the results. Lastly, in Section 6, we discuss mechanisms, and in Section 7, we conclude.

2 Setting and Experimental Design

2.1 Background

The Dominican Republic exemplifies the mathematics learning crisis facing much of Latin America and the Caribbean. In 2022, the country ranked third to last globally on the OECD's Programme for International Student Assessment (PISA), with 92 percent of students classified as low-performing in mathematics (OECD, 2023a,b). The consequences of these learning deficits become particularly acute during the transition to secondary education. The Dominican education system progresses from primary (grades 1-6) through lower secondary (grades 7-9) to upper secondary (grades 10-12), with students typically entering lower secondary around age 12.[3] Previous research indicates that up to 40 percent of students drop out between 8th and 12th grade (IDEICE, 2017).

Ninth grade represents a critical intervention point in this trajectory. Students at this stage have likely accumulated substantial learning deficits that threaten their ability to complete secondary education, yet they remain within the compulsory education system where remedial interventions can still be effectively implemented. Our study evaluates an intervention designed to address these challenges by providing targeted remedial mathematics training to ninth-grade students. To the extent that poor academic performance drives dropout decisions, improving learning outcomes may also promote educational retention; however, if students face binding non-academic constraints—such as economic pressures or family obligations—learning gains alone may prove insufficient to affect persistence. This study contributes to broader efforts to understand both pathways in the Dominican Republic and the wider LAC region.

2.2 Intervention

The study and intervention was implemented in partnership with the Ministry of Education of the Dominican Republic (MINERD) and the Abdul Latif Jameel Poverty Action Lab (J-PAL).[4] The intervention was implemented between October 2023 and May of 2024 in a sample of 38 9th grade classrooms in 9 schools.[5] The program had two core elements: a CAL intervention and a tutoring component. The first component introduced the use of a CAL platform (ALEKS) for two hours per week as part of schools' regular schedule, substituting two out of seven weekly hours scheduled for math instruction. Half of the classrooms were randomly selected to receive CAL licenses for all students and their math teachers (stratified by classroom size). The second component introduced small-group tutoring during extracurricular hours for approximately 20 percent of students in randomly selected classrooms from the first treatment group. Each component is summarized

[3] The school year runs from late August to late June, organized over two academic semesters. The 2023-24 school year officially started on August 28th 2023 and ended on June 21st 2024.

[4] The study is registered as AEARCTR-0012383 in the AEA registry, and was approved under protocol number 24-444 by the Teachers College IRB.

[5] Figure 1 shows a full timeline of the study and intervention.

in more detail below.

ALEKS platform. The program provided access to ALEKS, a math-focused CAL software. ALEKS uses predictive algorithms to deliver personalized learning experiences through adaptive assessments and assignments. These tools evaluate students' knowledge by identifying the math topics they master, the ones they do not, and the ones they are ready to learn. This approach is grounded in Knowledge Space Theory (KST), which posits that a subject's knowledge can be mapped as a set of solvable problems (Doignon and Falmagne, 1985; Falmagne et al., 1990). Accordingly, ALEKS assesses students' mastery by their ability to solve representative problems from each topic.

The software encompasses over 600 mathematics topics for ninth graders. Given the infeasibility of covering all topics within an academic year, the curriculum was narrowed to grade-appropriate content and essential prerequisites, yielding 337 topics organized into nine modules corresponding to content areas scheduled for instruction during the academic year.[6] Beyond practice exercises, ALEKS provides functionality for teachers to assign and evaluate homework and tests, along with a dashboard for tracking class and individual student progress.

All teachers assigned to a CAL classroom received an initial 2-hour training on the use of the learning technology. This training was provided by McGraw Hill, the developer of the learning platform. In addition to this, during the first 3 months of the program, teachers met one-on-one with a CAL aide for approximately one hour every two weeks. The CAL aide's role was to support them in accessing and monitoring students' assessments, tracking their progress, and identifying topics that students had not yet mastered for lesson planning. Importantly, one of the primary aspects of this extended training was to help teachers utilize the software's specific tools to ease their workload—including features that allow assigning and automatically grading homework—thus enabling teachers to save time that could potentially be devoted to other teaching activities.

Tutoring Intervention. The tutoring component explored complementarities between CAL and traditional instruction by providing up to three weekly hours of tutoring to academically at-risk students. In each classroom selected for tutoring, we targeted approximately 20 percent of students for the intervention. Sessions were managed by one or two tutors, depending on the number of tutees per classroom, which typically ranged from 5 to 8 students. To select tutoring recipients, mathematics teachers initially identified and ranked up to 10 students based on perceived academic need. To minimize selection bias, teachers were subsequently provided with baseline mathematics test results and given the opportunity to revise their selections. Approximately 70 percent of teachers modified their initial selections after receiving this test score information, and we used the final teacher selections to determine tutoring assignments.

Tutors were selected through a competitive process that included interviews. All tutors were either recent graduates or in the final stages of earning a Bachelor's degree in Mathematics with

[6] This selection was performed by two local experts on math pedagogy based on a review of the curriculum. The majority of prerequisite topics were identified by the learning software.

a focus on secondary education, and had experience in using information technology for teaching. Tutoring sessions were integrated into students' schedules during elective workshop periods, covering approximately three academic hours per week, typically divided into one to three weekly sessions.[7] Elective workshops are offered for all students attending extended-day schools in the Dominican Republic. These types of schools offer school days that start at 8am and end at 4pm, and the workshops typically cover areas outside of the core curriculum, such as computing or art lessons.

To ensure quality and consistency in the tutoring sessions, tutors were supported by a supervisor, who also had training and experience in mathematics education for high school students. The supervisor's role included reviewing individual session plans before each session, ensuring that the topics and instructional methods had been prepared before the session, and providing feedback to tutors if needed. In addition, tutors maintained close communication with the mathematics teacher to align their sessions with the classroom curriculum and were instructed to use the CAL platform for the tutoring sessions.[8] During the sessions, tutors reviewed topics covered in class during that week or revisited previous material, including pre-requisite topics that students found challenging. For instance, if a class was working on linear equations, a tutor may have reviewed the rules of signs before addressing the current topic.

The effectiveness of the tutoring sessions was monitored through unannounced visits. These visits were intended to identify potential issues or areas for improvement. One-on-one meetings between the supervisor and the observed tutor were arranged after these visits to provide feedback. Additionally, tutors completed weekly surveys reporting the number of tutoring hours conducted for each class and student to monitor attendance.

2.3 Sample and Randomization

The intervention was administered in 9 public high schools (with 38 ninth-grade classrooms) in the Dominican Republic. The sample size was constrained by schools' ability to meet the required technical and operational conditions. Schools were selected based on four criteria: (1) having an active computer lab, (2) having regular access to electricity and internet, (3) having an extended-day format,[9] and (4) having at least three ninth-grade classrooms. Additionally, we limited our sample to schools that had participated in *Programate*, a joint collaboration between MINERD and the World Bank that provided access to the ALEKS software to approximately 60 schools

[7] Out of 9 tutoring classrooms, 2 sections had one weekly meeting of 3 hours, 5 sections had two weekly meetings of 2 and then 1 hour, and 2 sections had three meetings for 1 hour each day.

[8] For example, tutors were encouraged to use the platform to identify topics where students were lagging behind and review these during tutoring sessions, or use the platform's accompanying exercises as practice. If a given session had a higher tutee/tutor ratio due to a tutor absence, the tutor was encouraged to alternate the use of CAL with small-group tutoring.

[9] Approximately 67 percent of public high schools in the Dominican Republic offered an extended school day format during the 2021-2022 school year (MINERD, 2023), where the school day starts at 8am and ends at 4pm.

between 2018 and 2023. This set of criteria yielded an initial list of 34 schools. Between April and August 2023, the research team, with support from MINERD, contacted these schools via phone to verify they met the criteria and assess their interest in participating. While all contacted schools expressed strong interest in participating, only 14 confirmed having reliable internet access and a functional computer lab.

Between September 4th and 15th, a member of the research team conducted school visits to validate the information collected during the Spring and Summer terms. Only 9 of the 14 visited schools had access to internet and functional computers. All participating schools were located in Santo Domingo and Santiago. Being the two largest cities in the country, they also have the largest concentration of students.[10]

Treatment was assigned using a two-stage randomization process at the classroom level. Initially, classrooms were ranked by size and paired with the closest matching classroom in terms of size. Within each pair, one classroom was randomly assigned to the treatment group and the other to the control group, ensuring similarly sized classrooms across groups. This pairing and randomization process was then repeated within the treatment group to assign half of these classrooms to receive the additional tutoring component: we created new pairs based on classroom size within the CAL group and randomly assigned the tutoring intervention within each pair. The study ultimately comprised 38 classrooms, with 19 assigned to receive the CAL software and the remaining 19 serving as controls.[11] Of the 19 CAL classrooms, 9 were further selected to receive group tutoring. The randomization design is summarized in Figure 2 and power calculations are included in Appendix A.1.

2.4 Implementation

The CAL intervention started on October 23rd and concluded on May 3rd.[12] During the first week of implementation, all treated students took a mandatory "Initial Knowledge Check" through the learning platform (this is the first activity in the platform). Based on the information from this initial assessment, the platform predicts the list of topics students master, do no yet master, and are ready to learn, and creates an individualized learning path for each student.

The first tutoring session took place on October 23rd and the last on April 26th. Tutoring sessions took place one to three days per week during extracurricular hours. In all but one school, tutoring sessions took place during hours allocated for optional workshops.[13] Optional workshops

[10] According to MINERD's Educational Statistics Yearbook for 2021-2022, approximately 45 percent of students were located in Santo Domingo and Santiago.

[11] Students in control classrooms received temporary licenses to access the software approximately 2 months after the endline data collection.

[12] This is equivalent to roughly 21 weeks, excluding winter, Holy Week, and two weeks where the use of the platform was interrupted due to training and data collection.

[13] One of the nine schools did not offer optional workshops, so students stayed after school hours to receive tutoring sessions.

were meant to cover extracurricular content and varied in scope. Some examples of the type of courses covered during these hours were computing, music or dance lessons. The workshops and tutoring sessions would typically take place between 2 and 3:30pm. In total, 61 students were selected to participate in these tutoring sessions.

3 Data

To evaluate the effectiveness of the program, we assess several key outcomes: students' academic performance, socioemotional skills, psychological well-being, and time use both inside and outside of school. Additionally, to better understand the potential drivers behind any observed effects, we examine changes in teachers' resource allocation based on student-reported information. These outcomes were collected using a combination of survey data, administrative records, data from the adaptive software, and brief math assessments. Each of these outcomes is described in detail below.

Student surveys and math assessments were administered at baseline, midline, and endline by the research team. Baseline surveys were collected in September 2023, followed by midline surveys in December and endline surveys in May 2024. Administrative records were collected and digitized after the end of the school year, between June and September 2024.

Academic performance. Our main outcome of interest is students' academic performance, as we want to examine whether the platform or tutoring component were effective in improving student learning in math. Assessments were designed using items of varying difficulty levels, adapted from MINERD's national assessments. Endline tests were designed by a pedagogical expert and followed the content and cognitive areas used on the 2019 Trends in International Mathematics and Science Study (TIMMS) 8th grade math assessments, which included seven content domains and three cognitive areas (Mullis and Martin, 2017).[14] We examine student performance in each of these content areas and cognitive domains, both at midline and endline. Moreover, we classify items according to their difficulty levels to assess differences in impacts by the grade level of the content.

In addition to the tests, between June and September 2024 we collected and digitized schools' administrative records, which were stored in secure rooms in each school. These records contained data on students' end-of-year school grades for math and Spanish (to test for adverse effects), their attendance records, whether they dropped out during the school year, and whether they passed or failed the grade.

Socioemotional skills. As students increasingly rely on digital devices for learning, there is a risk that socioemotional or non-cognitive skills may be negatively impacted. In baseline and endline surveys we measured the following three skills: perseverance, locus of control, and grit. Perseverance is measured through a real effort task adapted from Alan et al. (2019). In this task,

[14] The 8th grade TIMMS math assessments evaluated the following content domains: (1) Numbers, (2) Algebra, (3) Geometry, and (4) Data and Probability. The cognitive domains evaluated were: (1) Knowing, (2) Applying, and (3) Reasoning. TIMMS is only administered in the 4th and 8th grades.

students first answered a logic question, and then chose one of three options for their next step: attempting another question of the same difficulty, tackling a more difficult question, or giving up. Their choice is used as our measure of perseverance. To measure locus of control, or the extent to which students believe they have control over the outcome of events in their lives, we used an adaptation of Rotter's locus of control scale (Rotter, 1966). Lastly, we measured grit using a 4-item scale adapted from Carlana and La Ferrara (2021). We build indices for the grit and locus of control measures such that the minimum and maximum values are 0 and 1.

Student well-being. The nature of the learning platform, in which each student works independently, may have led to less opportunities for teamwork or support than regular instruction, potentially making students feel more isolated in class. Moreover, increased screen time has been linked to lower psychological well-being among teenagers (Stiglic and Viner, 2019; Twenge and Campbell, 2018). We used questions from the Children's Depression Screener (ChilD-S) developed by Frühe et al. (2012) to measure students' well-being.

Furthermore, by increasing students' math skills, or their awareness of their math skills, access to CAL may have mixed impacts on math anxiety. Following Araya et al. (2019), we created a math anxiety index using students' answers (on a Likert scale) to the following statements: 1) I get nervous before math tests and 2) I am worried of having bad grades in math.

Teacher behavior. The CAL platform may free up teachers' time by automating some of their daily tasks, such as developing and grading homework and tests. As a consequence, we may observe differences in the allocation of teachers' time and effort across classroom activities. To assess changes in these outcomes, we ask students about their perceptions of teacher behavior in the classroom (e.g. did the teacher give you additional support outside of the classroom?, did they clarify a lesson that you did not understand?) and their time management (e.g. did the teacher start class late?, did the teacher come to school?). We use their answers to build two indices of teacher behavior: 1) time mismanagement index and 2) teacher engagement index. The first index, on time mismanagement, is based on students' answers to a series of questions on teachers' timeliness to class and school. The second index is built upon students' answers to a set of questions around their engagement during class.[15]

CAL data. Used for both monitoring and assessment purposes. The software contains information on students' interactions with the software, including the number of sessions held, the date and duration of each session, and the number of newly mastered topics by each student. We use this information to assess compliance with the intervention and to measure progress within the treatment group. This data is available for the treatment group only.

[15] The questions used to build this index are: 1) Your math teacher keeps students engaged, 2) Asks questions to check for understanding, 3) Knows when everyone understands, 4) Makes every student work hard, 5) Assigns homework that helps you learn, 6) Reviews past topics, 7) Marks your work to help understanding, and 8) Wants you to explain your answers.

3.1 Balance Checks and Attrition

We present balance checks in Table 1 using information from baseline surveys. There are no statistically significant differences between the treatment and control groups across observable characteristics. Approximately 56 percent of the sample is female. The average age of students in our sample is close to 15 (the typical age for a students enrolled in 9th grade is 13 to 14 years old), reflecting the high levels of grade repetition in the country (approximately 19 percent of students in our sample repeated at least one grade). Baseline scores are relatively low: students answered an average of 6 out of 12 questions correctly. Moreover, we can observe that the average student in our sample comes from a low socioeconomic background; for instance, less than 40 percent of the sample has a mother or father who attended college.

As shown in Columns 1 and 5 of Table 2, survey and test completion rates were high at both midline and endline: 92 percent of the baseline sample completed surveys and tests at midline, and 90 percent did so at endline. The differences in completion rates between the control and treatment groups are small: 0.5 percentage points at midline and 2.9 percentage points at endline. While the latter difference is statistically significant at the 5 percent level, it is small enough not to raise substantial concerns.

Most importantly, the students who completed the surveys appear similar to those who did not. Columns 2-4 and 6-8 show minor differences in observable characteristics between respondents and non-respondents. For instance, older students were less likely to respond, with each additional year of age reducing the response probability by 3.7 percentage points; female students were about 4 percentage points more likely to respond than males; and students with higher baseline test scores were slightly more likely to remain in the sample (0.9 percentage points per SD). Overall, we believe the low rates of attrition and minimal differences between respondents and non-respondents do not compromise the integrity of our findings. As a robustness check, in Appendix Table A1, we present Lee (2009) bounds for the main performance outcomes.[16]

4 Empirical Framework

We estimate the impact of the program on student outcomes using the following regression:

$$Y_{ci} = \beta_0 + \beta_1 \text{CAL}_c + \beta_2 \text{Tutoring}_c + \beta_3 X_{ci} + \epsilon_{ci}$$

where Y_{ci} is the outcome of student i in classroom c and CAL_c is a binary variable that takes the value of 1 if classroom c was assigned to treatment and 0 otherwise. $Tutoring_c$ takes the value of 1 if the classroom was assigned to the group tutoring treatment and 0 otherwise. X_{ci} is a vector of

[16] We calculate bounds using a single regression with all three experimental groups ($y = \beta_1\text{CAL} + \beta_2\text{CAL} \times \text{Tutoring} + \varepsilon$). All groups are trimmed to equalize response rates at the minimum observed level (89.8%). The lower bound drops the highest-scoring students from higher-response groups (CAL and CAL+Tutoring); the upper bound drops the lowest-scoring students. This ensures treatment effect estimates are not biased by differential attrition.

baseline characteristics for student i in classroom c. In this specification, β_1 captures the average treatment effect (ATE) of the CAL software, while β_2 captures the additional impact ("value-added") of the tutoring component relative to classrooms that received only the CAL software. In the results section we present each coefficient separately and the sum of the two. The latter represents the effect for classrooms assigned to both the CAL and tutoring interventions, relative to the control group.

Throughout this paper, we use randomization-based inference as our preferred methodology for analysis, which has gained prominence over traditional sampling-based methods in recent years (Athey and Imbens, 2017; Green et al., 2011; Young, 2019). This approach leverages the experimental design to derive statistical inferences directly from the randomization procedure, ensuring that our methods align closely with the study's design. Randomization inference is particularly advantageous in the context of nonstandard randomization techniques, such as the pairwise randomization used here, as it allows us to account for the specific structure of the randomization. Moreover, it enables us to obtain nearly exact p-values without relying on assumptions about the shape of the sampling distribution, which is especially helpful in small samples or when the number of clusters is limited, as is the case in this study. In such situations, traditional assumptions about the sampling distribution may be less reliable, making randomization inference a more robust approach.

The validity of our findings relies on the "no-interference" or Stable Unit Treatment Value Assumption (Rubin, 1978), which assumes that the treatment of one classroom does not affect the potential outcomes of students in another. The platform used in this study required each student to log in using a unique username and password, and its use was closely monitored during the designated period for this task. These measures minimize the likelihood that students in non-treated classrooms could have accessed or benefited from the platform, thereby supporting the assumption of no interference. However, it is conceivable that treated students may have interacted with control students in ways that influenced their outcomes (e.g., through assistance with homework or exam preparation). Additionally, the assignment of a classroom to treatment may have impacted how teachers interacted with students in non-treated classrooms. We find no evidence of such spillovers: discussions with the teaching team and focus groups with students revealed no indication that teachers modified their instruction in control classrooms in response to the treatment, nor that treated students collaborated with peers across classroom boundaries.

5 Results

5.1 Tutoring and CAL Use

The results from the initial CAL assessment suggest that, at the time of completion, the median student mastered only 2.7 percent (9) of the 337 math topics included as part of the course.

Throughout the school year, the median treated student spent approximately 14 hours (approx. 38 minutes per week) on the platform and learned an average of 45 new math topics through it. As shown in Figures 3 and 4, the distribution of the number of hours and the number of topics learned on the platform is left-skewed, with the top 10th percentile of students spending over 38 hours using the platform and learning more than 188 new math topics throughout the school year.

To assess out-of-school usage, we measure the share of total platform time occurring on non-scheduled days. The median student used the platform outside school for 15 percent of their total time, or approximately two hours. This modest figure suggests that observed effects reflect a reallocation of in-class time rather than additional instruction.

Of the 61 students assigned to the tutoring treatment arm, all received at least 2 hours of tutoring, with the median student receiving 29 hours, or approximately 1.4 academic hours per week. Note that this is significantly below the intended dosage of 3 hours per week. This is partially explained by student absenteeism during the tutoring sessions. On a given week, up to 40 percent of students failed to show up to at least one of the meetings, based on monitoring data. We include a more detailed discussion of this in Section 6.

5.2 Student Performance

We present Intention-to-Treat (ITT) effects roughly two (midline) and six months (endline) after the start of the program in Table 3. Columns 1 and 2 present results at midline and columns 3 and 4 at endline. We find that by midline the CAL-only treatment had produced gains of approximately 0.17-0.21 standard deviations, as shown in Columns 1 and 2. Although large, these results are not statistically significant. These impacts are roughly consistent across specifications (with or without covariates). We do not find significant effects from the tutoring treatment at midline, likely because students had only received 8 tutoring sessions at this stage.

By endline, CAL-only effects had increased to 0.28-0.31 standard deviations (statistically significant at the 5 percent level), as shown in Columns 3 and 4. The coefficient associated with the tutoring add-on (β_2) was negative (-0.19 SD with covariates), but not statistically distinguishable from zero. The combined effect of CAL plus tutoring relative to control ($\beta_1 + \beta_2 = 0.09$ SD) was also not statistically significant. We cannot reject that the tutoring component had zero effect, though the point estimates suggest the possibility of interference between interventions. We find that students in tutoring classrooms—including non-tutees—exhibited lower platform engagement (Figure 9) and reduced locus of control (Appendix Table B8) relative to the CAL-only group, suggesting that the presence of tutors may have affected student behavior in ways that undermined the effectiveness of the adaptive technology. We explore these mechanisms in detail in Section 6. Effects on performance from the CAL component align with those found in other work using computer-assisted instruction in low or middle income countries. For example, Araya et al. (2019) report effects of 0.21-0.27 standard deviations (SD) from a computer-assisted, game-based math learning platform in Chile after seven months of program exposure. In India, Muralidharan et al. (2019) find effects of 0.37 SD after 4.5 months. Finally, Büchel et al. (2022) observe effects of up to 0.17 SD from a computer-assisted learning software in El Salvador over a six-month period.

5.3 Heterogeneity in Program Effects

As students progressed through the software, they were targeted with exercises that catered to their learning levels. Following Muralidharan et al. (2019), we classify test topics into two categories–below and at grade level–and test whether students fare differently across these two categories in the tests. Table 4 suggests that the CAL treatment was effective in improving performance on both below and at grade level content, with impacts ranging from 0.21 to 0.28 standard deviations. This pattern differs from that found in Muralidharan et al. (2019), which finds large impacts on below grade math items but no impacts on at grade level items.[17] However, the patterns are consistent with what we would expect by looking at software use, as the software was designed to cover both below and at grade level content. At midline, when students had spent most time on modules covering below grade level topics, we find that effects were largely concentrated on test items covering this type of content.

In Table 5, we present effects across content areas included in TIMMS mathematics assessments for 8th grade. We observe positive differences from the CAL treatment on the algebra, numbers, and geometry content areas by endline, although the coefficient for geometry is not statistically significant. We do not find impacts on data and probability, the one TIMMS content area that was not covered in the platform or in the Dominican Republic's 9th grade curriculum. For the tutoring component, results seem to mirror those found on Table 3, as classrooms assigned to a tutor show negative but statistically insignificant differences, when compared to the CAL-only treatment, across all content areas except for data and probability. Beyond content areas, TIMSS also evaluates students' abilities in three cognitive domains: knowing, applying, and reasoning. In Table 6, we present results across these cognitive areas. As shown, we find large and positive effects on all cognitive areas at endline, ranging from 0.16 to 0.31 standard deviations, although results are not statistically significant for the "applying" cognitive area.

Lastly, Figure 6 presents the distribution of performance for both the treatment and control groups across baseline performance percentiles. Although confidence intervals overlap at several points, we observe that the treatment group consistently outperformed the control group at all levels of baseline performance.

5.4 Secondary Outcomes

Beyond academic performance, we examine whether the intervention affected student retention, motivation, socioemotional skills, psychological well-being, beliefs about own performance, and teacher

[17] In Hindi, the authors find significant effects for questions at and below grade level.

behavior. Most secondary outcomes show no statistically significant effects, with four notable exceptions: greater overconfidence in anticipated test performance and lower locus of control among CAL+Tutoring students, increased preference for easier tasks on a real-effort persistence measure under CAL, and reduced teacher time mismanagement under CAL+Tutoring. We summarize these findings briefly here; full results are reported in Appendix B.

Student Retention and Motivation. Despite the substantial learning gains from the CAL intervention, we find no effects on dropout rates, school absenteeism, intrinsic motivation, or stated preferences for mathematics. The null effect on dropout is consistent with the possibility that students in this context face binding non-academic constraints—such as economic pressures or family obligations—that learning gains alone cannot address.

Student Beliefs and Expectations. We examine whether the interventions affected students' calibration of their own academic performance. At endline, students were asked to predict their test score prior to taking the assessment, allowing us to construct a measure of overconfidence as the difference between predicted and actual performance. As shown in Table 7, students in CAL+Tutoring classrooms exhibited marginally higher overconfidence than the control group (5.9 percentage points, $p < 0.10$ with covariates), while CAL-only students showed no significant difference. Figure 5 illustrates these patterns: while actual performance was highest in CAL-only classrooms, expected performance was similar across groups, resulting in greater overconfidence among CAL+Tutoring students. We explore the implications of these belief patterns for platform engagement in Subsection 6.4.

Socioemotional Skills. We examine effects on self-efficacy, grit, perseverance, and locus of control. The CAL intervention had no significant impact on self-efficacy or grit. However, students in CAL-only classrooms were more likely to choose an easier task over a more challenging alternative in the perseverance exercise, suggesting that the adaptive nature of the platform may have reduced willingness to take on difficult problems. Notably, the tutoring component produced a small but statistically significant negative effect on locus of control, indicating that students in the tutoring group were more likely to attribute outcomes to external factors rather than their own effort.

Psychological Well-Being. We find no effects on a depression index based on the Children's Depression Screener. Students in CAL-only classrooms reported higher math anxiety (nervousness before tests) relative to both the control and CAL+Tutoring groups. While math anxiety is typically associated with lower performance, this finding may reflect heightened awareness of—and engagement with—their mathematical development.

Student Time Use. One potential concern is that CAL might displace other productive activities or increase screen time outside of school. We find no significant effects on students' time allocation across eight categories: in-person classes, unpaid work at home, unpaid work outside the home, paid work, studying outside of school, meeting friends, entertainment, and sports. This suggests that the intervention operated primarily through the substitution of regular math instruction rather than through changes in how students allocated time outside of CAL sessions.

Teacher Behavior. We find that classrooms assigned to CAL+Tutoring showed significantly lower teacher time mismanagement (0.23 SD, $p<0.05$), indicating that teachers in these classrooms were more punctual and had fewer absences. This may reflect increased accountability from the presence of tutors or greater organizational structure required by the combined intervention. We do not find significant effects on a teacher engagement index, though CAL-only classrooms showed marginally lower engagement ($p<0.10$ without controls), possibly because the platform substitutes for some direct instruction.

6 Mechanisms

The results of the experiment indicate that the impacts of the CAL intervention depend critically on students' engagement with the platform and on how the intervention interacts with existing instructional practices. In this section, we first rule out differential implementation as an explanation for the divergent effects between treatment arms, and then synthesize quantitative evidence from platform usage, student beliefs, and performance outcomes, together with qualitative findings from focus groups, to discuss potential mechanisms.

6.1 Implementation Fidelity

To assess whether differential implementation could explain the divergent effects, we collected weekly monitoring data on session delivery across all 19 CAL classrooms over 21 weeks.[18] Overall, CAL sessions achieved 67.7 percent of expected dosage (28.4 of 42 planned sessions). Fidelity did not differ between classrooms assigned to CAL only versus CAL plus tutoring: the former achieved 68.8 percent compliance while the latter achieved 66.4 percent. Among the nine classrooms receiving both components, tutoring achieved 63.5 percent of expected dosage-statistically indistinguishable from CAL compliance in those same classrooms.

When sessions were cancelled, the reasons were predominantly exogenous. Among cancelled CAL sessions, 74.5 percent were due to school closures, institutional activities, or infrastructure failures; similarly, 75.5 percent of cancelled tutoring sessions were attributable to such factors. The remaining cancellations reflected intervention-specific barriers: laboratory access and ALEKS

[18] Compliance rates are calculated at the classroom level (sessions held divided by sessions scheduled).

licensing issues for CAL, and tutor availability and space constraints for tutoring. The similar compliance rates and barrier distributions across components rule out differential implementation as an explanation for the attenuated effects in the CAL plus tutoring arm.

6.2 Engagement Intensity and Dose–Response

A key mechanism through which the CAL intervention appears to affect learning is sustained engagement with adaptive, self-directed practice. Platform data reveal substantial heterogeneity in usage: the median treated student used ALEKS for approximately 14 hours over the school year and mastered 45 topics, while many others used the platform for over 80 hours, reaching mastery of over 300 topics (see Figures 3 and 4). This variation is economically meaningful. Using random assignment as an instrument for hours of platform use, we estimate that each additional hour of CAL engagement increases endline test scores by approximately 0.026–0.027 standard deviations (see Table 8). The estimated dose–response relationship exhibits diminishing returns, with gains increasing steeply up to roughly 1500 minutes (25 hours) of use before flattening (see Figure A2). In Table 8 we extrapolate our estimates to 1500 minutes of use, which, assuming 80 percent attendance, would be equivalent to roughly 5 months of implementation. Our results suggest that implementing the program with fidelity for 5 months would lead to test score gains of 0.73 standard deviations.[19]

While extrapolation beyond observed usage should be interpreted cautiously, these estimates suggest that higher-fidelity implementation could plausibly generate larger learning gains. Implementation frictions, including connectivity issues, limited device availability, and disruptions caused by school activities, reduced effective exposure to the platform and may therefore have attenuated impacts relative to this benchmark.

6.3 Substitution of Effort and Classroom-Level Spillovers

Despite similar access to the platform, students in classrooms assigned to CAL+tutoring spent less time on ALEKS and mastered fewer topics than students in CAL-only classrooms. We also observe that students in the tutoring arm put less effort while using the platform.[20] Importantly, this pattern is observed not only among students selected for tutoring, but also among their non-tutored classmates, indicating that the tutoring intervention affected classroom-level learning dynamics. Figure 9 shows that the average student assigned to a CAL-only classroom spent a total of 1232 minutes using the platform (56 minutes per week) while the average student in a CAL+Tutoring classroom spent 1082 minutes (49 minutes per week). We use the relationship between minutes of use and test scores to predict how this difference in use may affect results. Assuming a linear dose-response relationship between the two, we find that this difference in time use may account

[19]This extrapolation requires the assumption of heterogeneity in treatment effects. As shown in Figure 6, we do not find evidence of effect heterogeneity across baseline performance.

[20]We define effort as the ratio of topics learned to time spent on the platform.

for approximately 34 percent (0.07 SD) of the observed difference in endline performance between classes with tutoring and those that only included the CAL component (see Table 8). Within CAL+Tutoring classrooms, both non-tutees and tutees had considerably lower levels of use than the average student in a CAL-only classroom, but the difference in use was significantly lower for tutees, who spent only 1000 minutes (or 45 minutes per week) using the platform.

One possible interpretation is that tutoring altered how students allocated learning effort. Qualitative evidence suggests that students in tutoring classrooms often relied on tutors' explanations to resolve difficulties; this may have affected their ability to persist independently through adaptive exercises. Although tutoring dosage fell short of intended levels due to absenteeism and scheduling disruptions, the presence of tutors may nonetheless have provided an alternative pathway to understanding that substituted for self-directed platform use.

6.4 Agency, Beliefs, and Attribution

Differences in engagement across treatment arms are accompanied by differences in students' beliefs. Survey data indicate that students in CAL-only classrooms formed more accurate expectations about their academic performance, while students in CAL plus tutoring classrooms exhibited higher overconfidence and a statistically significant reduction in locus of control (see Tables 7 and B8). These patterns suggest that having been assigned a tutor may have shifted attribution of learning away from individual effort and toward external support.

Platform design features—such as difficulty jumps, loss of progress after errors, and limited explanatory feedback—likely increased the cost of persistence for some students, as highlighted in focus group discussions. In CAL-only classrooms, students had limited alternatives and may therefore have been more likely to persevere despite these frictions. In tutoring classrooms, increased access to human explanations may have reduced tolerance for platform-related challenges, reinforcing disengagement and external attribution. While these mechanisms cannot be conclusively established, they are consistent with the observed patterns in engagement and beliefs.

6.5 Implications

Taken together, the evidence suggests that the lack of additional gains from the tutoring add-on reflects more than implementation shortfalls alone. Rather, tutoring appears to have interacted with the CAL intervention in ways that weakened the mechanisms through which the technology generates learning gains—by reducing platform engagement and altering students' beliefs about effort and learning. These findings highlight the importance of considering behavioral responses when combining educational interventions. Adaptive learning technologies may be most effective when they preserve student agency and sustained engagement. When layered with interventions that substitute for these processes, and particularly in settings with implementation frictions, their effectiveness may be diminished. More broadly, the results underscore the value of empirically testing intervention bundles rather than assuming additive effects.

7 Conclusion

This paper contributes to the growing literature on computer-assisted learning by testing an innovative intervention that replaces nearly one-third of regular math instruction with the use of an adaptive learning platform. Conducted in the Dominican Republic, a lower-middle-income country, this study highlights both the challenges and transformative potential of introducing technology-based tools in public school classrooms. We find that substituting two weekly hours of regular instruction with CAL use significantly improves student performance on math tests by up to 0.3 standard deviations, with these gains being consistent across various difficulty levels and subject areas.

When CAL was combined with small-group tutoring, we did not observe the additive effects one might expect from layering two interventions. Point estimates suggest that classrooms receiving both CAL and tutoring performed no better—and possibly worse—than those receiving CAL alone, though this difference is not statistically significant. These findings suggest that combining interventions does not guarantee improved outcomes, and that successful implementation requires attention to how different components interact. Reduced platform usage and effort among students in the combined treatment group partially account for this attenuation, though we cannot conclusively identify why engagement declined. We do, however, document effects on locus of control and wellbeing: relative to CAL-only students, those in the combined treatment exhibited lower locus of control and lower math anxiety. Together, these results underscore the importance of understanding how technology-based interventions interact with complementary classroom supports.

Overall, this paper demonstrates the substantial potential of technology to enhance education systems worldwide. However, it also highlights the critical importance of understanding the dynamics between technology integration and concurrent interventions, such as tutoring, within the school context. Furthermore, we underscore the importance of infrastructure and logistical planning in the successful implementation of school-based technology programs, emphasizing the need for continuous monitoring and adaptive strategies to minimize disruptions. More broadly, as artificial intelligence becomes increasingly integrated into education systems worldwide, understanding how to effectively combine human and technological inputs will become essential. This study provides crucial evidence that such integration requires careful attention to behavioral responses.

References

Alan, S., T. Boneva, and S. Ertac (2019). Ever failed, try again, succeed better: Results from a randomized educational intervention on grit. *The Quarterly Journal of Economics 134*(3), 1121–1162.

Angel-Urdinola, D., C. Avitabile, and M. Chinen (2023). Can digital personalized learning for mathematics remediation level the playing field in higher education? experimental evidence from ecuador.

Angrist, J. and V. Lavy (2002). New evidence on classroom computers and pupil learning. *The Economic Journal 112*(482), 735–765.

Araya, R., E. A. Ortiz, N. L. Bottan, and J. P. Cristia (2019). Does gamification in education work?: Experimental evidence from chile.

Athey, S. and G. W. Imbens (2017). The econometrics of randomized experiments. In *Handbook of economic field experiments*, Volume 1, pp. 73–140. Elsevier.

Augustine, C. H., J. S. McCombs, J. F. Pane, H. L. Schwartz, J. Schweig, A. McEachin, and K. Siler-Evans (2016). *Learning from Summer: Effects of Voluntary Summer Learning Programs on Low-Income Urban Youth. RAND Summer Learning Series. Research Report. RR-1557-WF.* ERIC.

Barrow, L., L. Markman, and C. E. Rouse (2009). Technology's edge: The educational benefits of computer-aided instruction. *American Economic Journal: Economic Policy 1*(1), 52–74.

Bloom, B. S. (1984). The 2 sigma problem: The search for methods of group instruction as effective as one-to-one tutoring. *Educational researcher 13*(6), 4–16.

Büchel, K., M. Jakob, C. Kühnhanss, D. Steffen, and A. Brunetti (2022). The relative effectiveness of teachers and learning software: Evidence from a field experiment in el salvador. *Journal of labor economics 40*(3), 737–777.

Carlana, M. and E. La Ferrara (2021). Apart but connected: Online tutoring and student outcomes during the covid-19 pandemic.

Doignon, J. P. and J. C. Falmagne (1985). Spaces for the assessment of knowledge. *International Journal of Man-Machine Studies 23*(2), 175–196.

Dowker, A., A. Sarkar, and C. Y. Looi (2016). Mathematics anxiety: What have we learned in 60 years? *Frontiers in psychology 7*, 508.

Duckworth, A. L., C. Peterson, M. D. Matthews, and D. R. Kelly (2007). Grit: perseverance and passion for long-term goals. *Journal of personality and social psychology 92*(6), 1087.

Evans, D. K. and F. Yuan (2022). How big are effect sizes in international education studies? *Educational Evaluation and Policy Analysis 44*(3), 532–540.

Falmagne, J.-C., M. Koppen, M. Villano, J.-P. Doignon, and L. Johannesen (1990). Introduction to knowledge spaces: How to build, test, and search them. *Psychological Review 97*(2), 201–224.

Frühe, B., A.-K. Allgaier, K. Pietsch, M. Baethmann, J. Peters, S. Kellnar, A. Heep, S. Burdach, D. von Schweinitz, and G. Schulte-Körne (2012). Children's depression screener (child-s): development and validation of a depression screening instrument for children in pediatric care. *Child Psychiatry & Human Development 43*, 137–151.

Glewwe, P. and K. Muralidharan (2016). Improving education outcomes in developing countries: Evidence, knowledge gaps, and policy implications. In *Handbook of the Economics of Education*, Volume 5, pp. 653–743. Elsevier.

Green, D. P., P. M. Aronow, D. E. Bergan, P. Greene, C. Paris, and B. I. Weinberger (2011). Does knowledge of constitutional principles increase support for civil liberties? Results from a randomized field experiment. *The Journal of Politics 73*(2), 463–476.

Guryan, J., J. Ludwig, M. P. Bhatt, P. J. Cook, J. M. Davis, K. Dodge, G. Farkas, R. G. Fryer Jr, S. Mayer, H. Pollack, et al. (2023). Not too late: Improving academic outcomes among adolescents. *American Economic Review 113*(3), 738–765.

Instituto Dominicano de Evaluación e Investigación de la Calidad Educativa (IDEICE) (2017). Title of the document. Accessed: 2023-04-28.

Interamerican Development Bank (2023). Latin America and the Caribbean in PISA 2022: How did the Region Perform? Technical report, Interamerican Development Bank.

Kraft, M. A. and G. T. Falken (2020). A blueprint for scaling tutoring across public schools. edworkingpaper no. 20-335. *Annenberg Institute for School Reform at Brown University*.

Kraft, M. A., B. E. Schueler, and G. Falken (2024, August). What impacts should we expect from tutoring at scale? exploring meta-analytic generalizability.

Lee, D. S. (2009). Training, wages, and sample selection: Estimating sharp bounds on treatment effects. *The Review of Economic Studies 76*(3), 1071–1102.

Linden, L., A. Banerjee, and E. Duflo (2003). Computer-assisted learning: Evidence from a randomized experiment. *Poverty Action Lab Paper 5*.

Ministerio de Educación de la República Dominicana (MINERD) (2023, March). Title of the document. Accessed: 2023-04-28.

Mullis, I. V. and M. O. Martin (2017). *TIMSS 2019 Assessment Frameworks.* ERIC.

Muralidharan, K. and A. Singh (2023, 11). Improving schooling productivity through computer-aided instruction:Experimental evidence from Rajasthan*.

Muralidharan, K. and A. Singh (2025). Improving schooling productivity through computer-aided instruction: Experimental evidence from rajasthan. *American Economic Review*. Forthcoming.

Muralidharan, K., A. Singh, and A. J. Ganimian (2019). Disrupting education? experimental evidence on technology-aided instruction in india. *American Economic Review 109*(4), 1426–60.

Nickow, A., P. Oreopoulos, and V. Quan (2020). The impressive effects of tutoring on prek-12 learning: A systematic review and meta-analysis of the experimental evidence.

OECD (2023a). *PISA 2022 Results (Volume I).*

OECD (2023b). *PISA 2022 Results (Volume II).*

Oreopoulos, P., O. Keyes-Krysakowski, and D. Agarwal (2026, January). How in-school supervised ed-tech support produces massive learning gains: A khan academy field experiment in india. Working Paper 34683, National Bureau of Economic Research.

Rotter, J. B. (1966). Generalized expectancies for internal versus external control of reinforcement. *Psychological monographs: General and applied 80*(1), 1.

Rubin, D. B. (1978). Bayesian inference for causal effects: The role of randomization. *The Annals of statistics*, 34–58.

Stiglic, N. and R. M. Viner (2019). Effects of screentime on the health and well-being of children and adolescents: a systematic review of reviews. *BMJ open 9*(1), e023191.

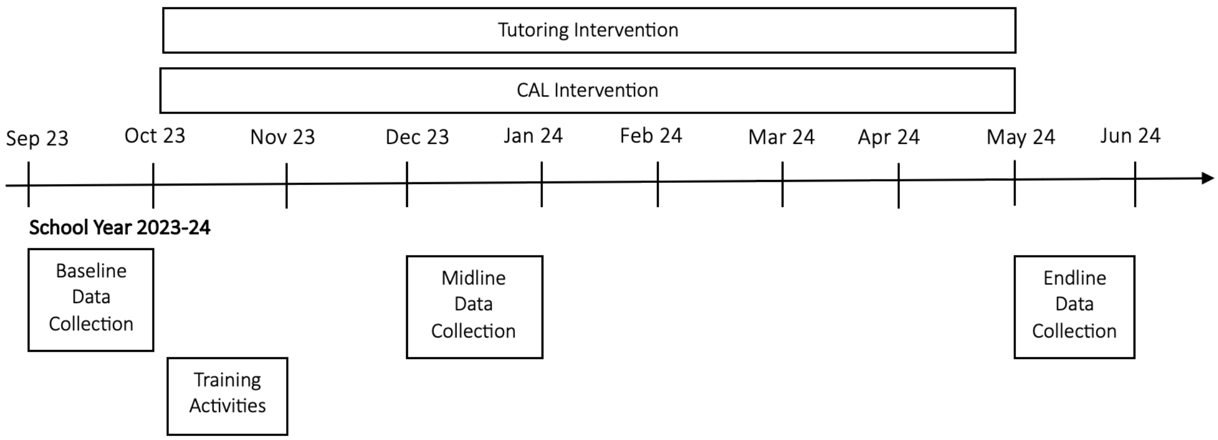
Twenge, J. M. and W. K. Campbell (2018). Associations between screen time and lower psychological well-being among children and adolescents: Evidence from a population-based study. *Preventive medicine reports 12*, 271–283.

World Bank (2017). *World development report 2018: Learning to realize education's promise.* The World Bank.

Young, A. (2019). Channeling fisher: Randomization tests and the statistical insignificance of seemingly significant experimental results. *The quarterly journal of economics 134*(2), 557–598.

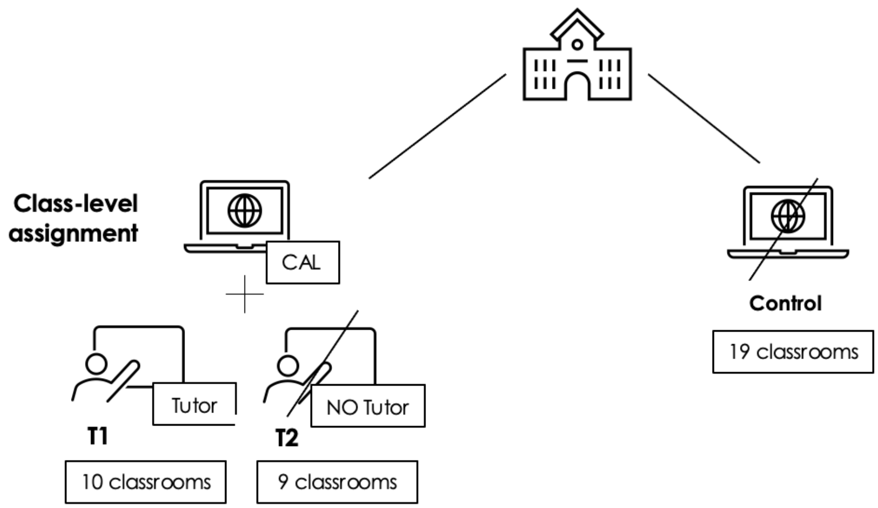
Figures and Tables

Figure 1: Timeline for Intervention and Study



Notes: The CAL intervention ran from October 23, 2023 to May 3, 2024 (approximately 21 instructional weeks). The tutoring intervention ran from October 23, 2023 to April 26, 2024. Platform use was interrupted during winter break, Holy Week, and two weeks allocated for training and data collection. Baseline surveys were administered in September 2023, midline surveys in December 2023, and endline surveys in May 2024

Figure 2: Randomization Design



Notes: T2 is composed of 10 classrooms assigned to receive access to the CAL software. T1 is composed of 9 classrooms where in addition to receiving access to the software, the bottom quarter of students, in terms of performance, were to receive tutoring sessions in small groups during extracurricular hours. The control group, containing 19 classrooms, did not receive access to the CAL or tutoring components.

Figure 3: Time Spent on CAL

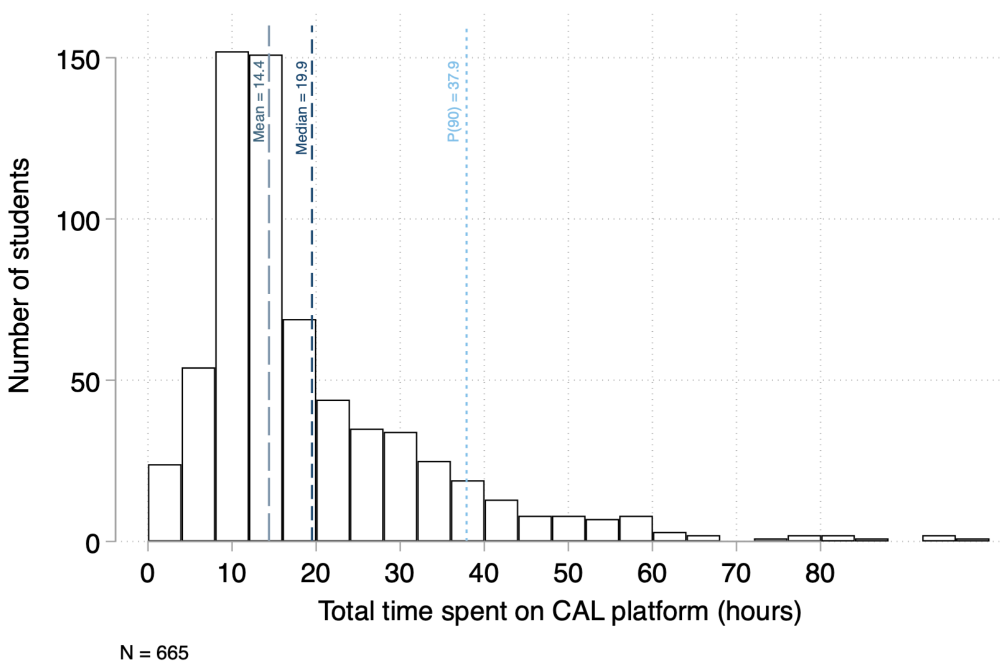


Figure 4: Topics Learned on CAL

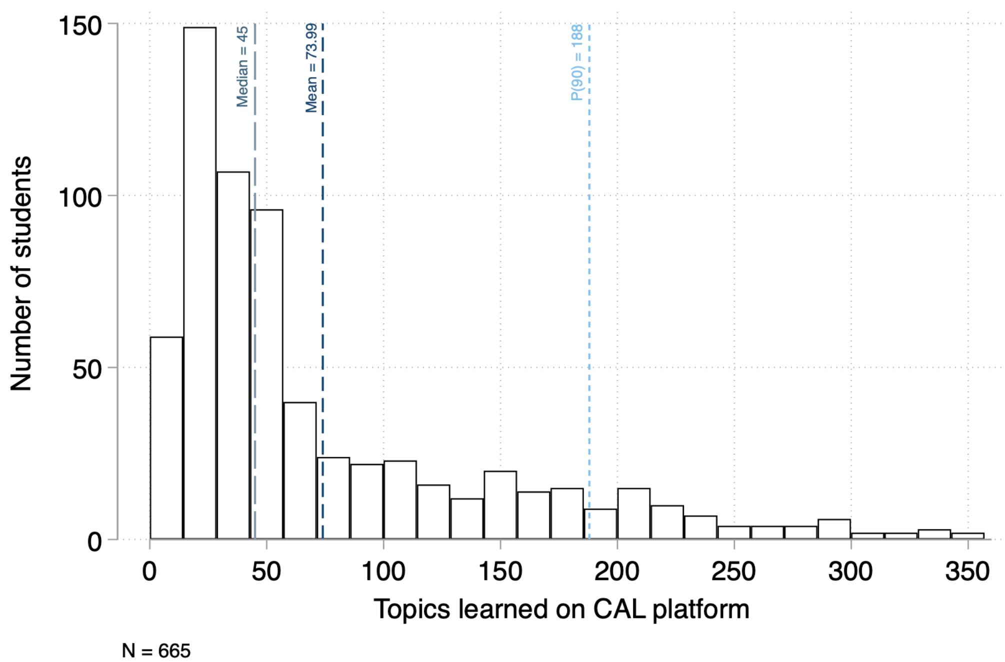
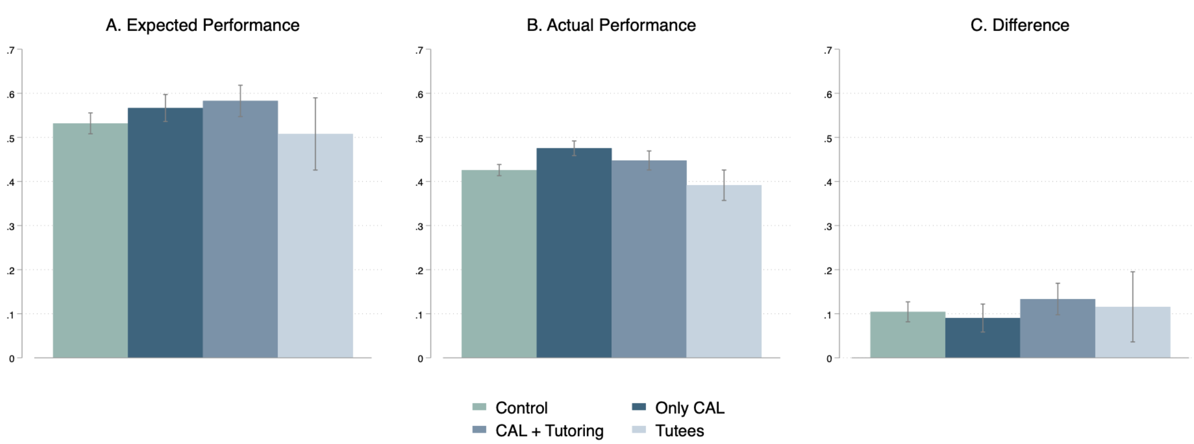
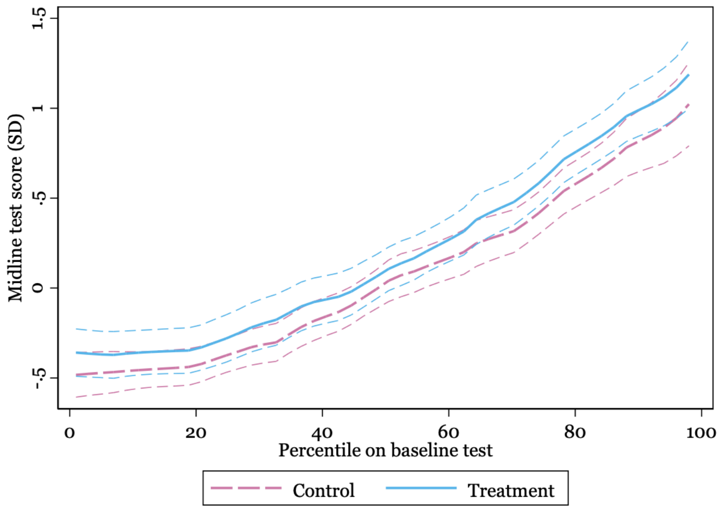


Figure 5: Expected Performance on Endline Math Tests



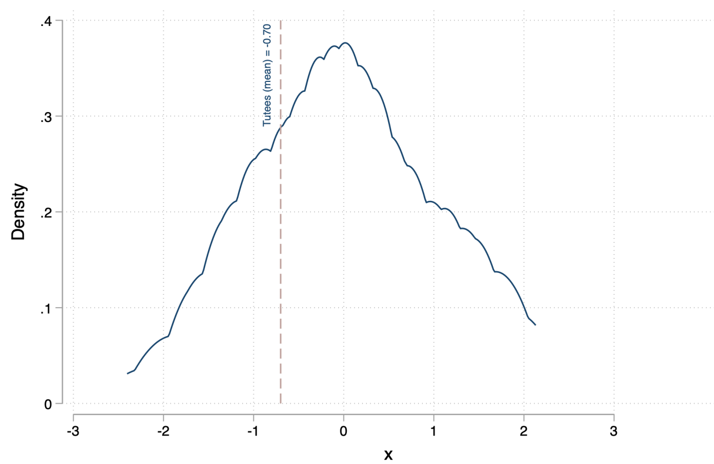
Notes: The CAL+Tutoring group excludes tutees. Panel C shows the difference between students' reported expected performance (share of correct answers) and their actual performance.

Figure 6: Non-Parametric Estimates of Treatment Effects by Baseline Performance



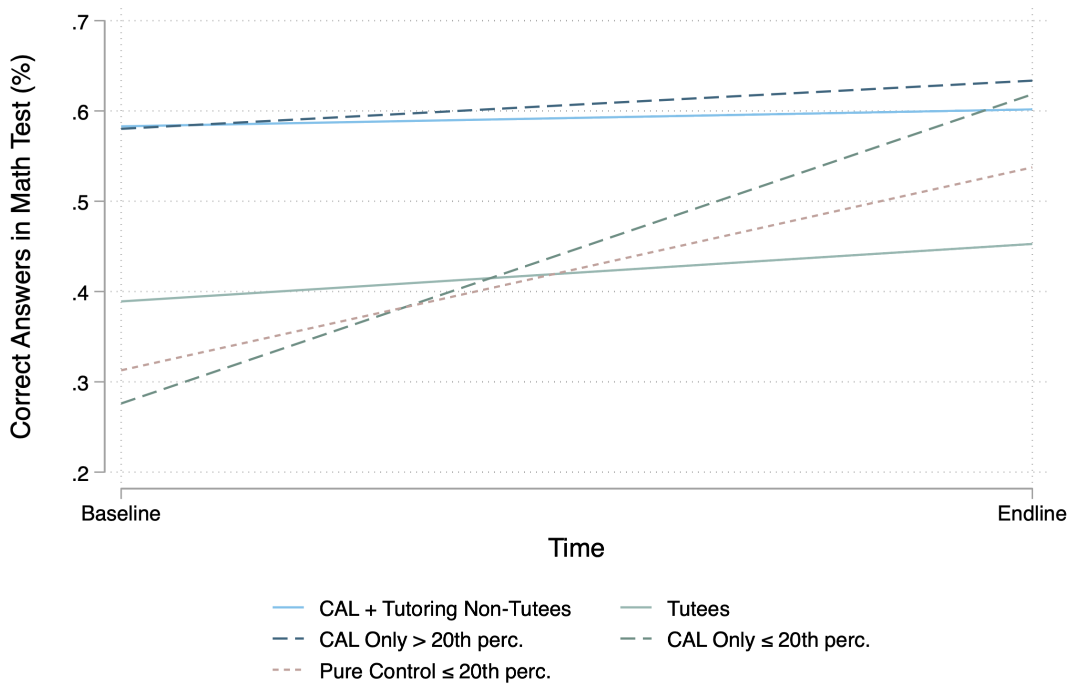
Notes: Following Muralidharan et al. (2019), the figure presents kernel-weighted local mean smoothed plots relating endline test scores to percentiles in baseline tests, separately for the treatment and control groups. The dashed lines show 95% confidence intervals.

Figure 7: Selection of Tutees and Distribution of Performance



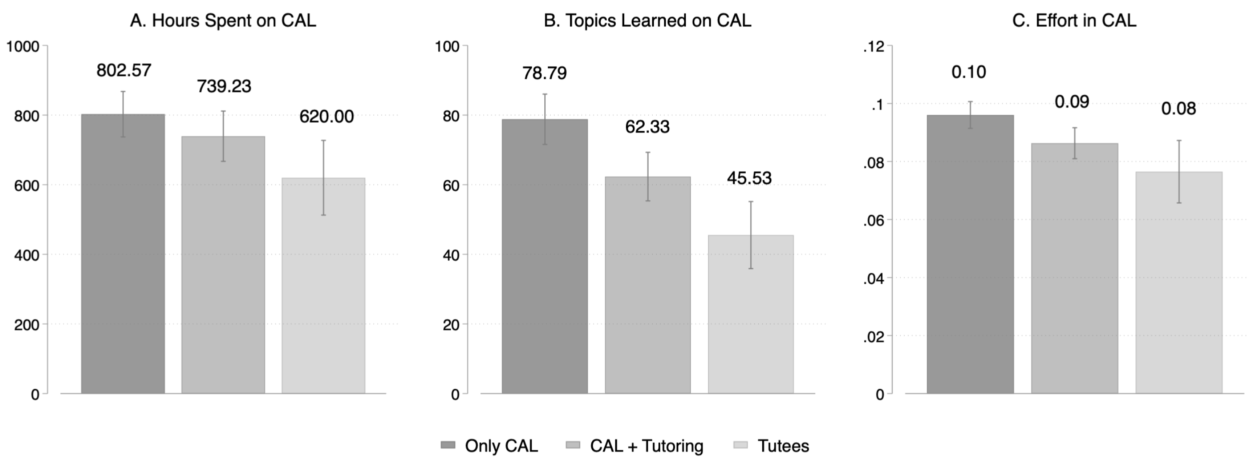
Notes: The blue line shows the average baseline score for selected tutees.

Figure 8: Share of Correct Answers by Performance Groups



Notes: The 20th percentile or below threshold of performance is used as a counterfactual for tutees. "CAL+Tutoring Non-Tutees" shows trends between baseline and endline for students who did not directly receive the group tutoring intervention but were in classrooms where approx. 20 percent of students did. "Tutees" are students who were assigned to the tutoring intervention in selected classrooms. "CAL-Only $>$ 20th perc." are students in CAL-only classrooms who are above the 20th percentile of performance. "CAL Only $\leq$ 20th percentile" are students in CAL-only classrooms who were below the 20th percentile of performance. "Pure Control $\leq$ 20th percentile" are students below the 20th percentile performance threshold in control classrooms. Dashed lines are approximate counterfactual trends for students in group tutoring classrooms who did not receive tutoring (blue) and students in tutoring classrooms who did receive tutoring (green and pink).

Figure 9: Platform Use by Groups



Notes: The CAL+Tutoring group excludes tutees.

Table 1: Balance Table

	Control		Treatment			
Variable	N	Mean/(SE)	N	Mean/(SE)	N	Difference
Female	669	0.552	664	0.566	1333	-0.015
		(0.019)		(0.019)		
Age	615	14.525	629	14.458	1244	0.067
		(0.033)		(0.033)		
Math test scores	620	6.281	631	6.442	1251	-0.162
		(0.108)		(0.110)		
Mother went to college	622	0.293	636	0.319	1258	-0.027
		(0.018)		(0.018)		
Father went to college	622	0.186	636	0.212	1258	-0.026
		(0.016)		(0.016)		
Mother went to high school	622	0.693	636	0.708	1258	-0.015
		(0.019)		(0.018)		
Father went to high school	622	0.564	636	0.558	1258	0.006
		(0.020)		(0.020)		
F-test of joint significance						0.635
F-test, number of observations						1237

Notes: *** p<0.01, ** p<0.05, * p<0.1. Math test scores are based on a short assessment administered at baseline.

Table 2: Attrition between Baseline, Midline, and Endline

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Midline Survey				Endline Survey			
	All	All	Treat	Control	All	All	Treat	Control
Treatment	0.005	0.013			0.029*	0.026		
	(0.015)	(0.014)			(0.015)	(0.016)		
Student's age		-0.009	-0.022	0.007		-0.025**	-0.034**	-0.018
		(0.011)	(0.014)	(0.017)		(0.012)	(0.015)	(0.020)
Student is female		-0.028*	-0.020	-0.034		0.008	0.036	-0.013
		(0.015)	(0.021)	(0.023)		(0.017)	(0.022)	(0.026)
Father is unemployed		-0.024	-0.036	-0.002		-0.002	0.036	-0.027
		(0.027)	(0.038)	(0.039)		(0.030)	(0.040)	(0.045)
Mother is unemployed		-0.016	0.013	-0.046*		0.001	0.019	-0.012
		(0.017)	(0.023)	(0.024)		(0.019)	(0.025)	(0.028)
Baseline test score		0.003	0.002	0.005		0.008**	0.011**	0.005
		(0.003)	(0.004)	(0.004)		(0.003)	(0.004)	(0.005)
Repeated a school year		-0.011	0.024	-0.043		-0.005	0.036	-0.038
		(0.024)	(0.033)	(0.036)		(0.027)	(0.035)	(0.042)
Mother went to college		0.019	0.023	0.018		-0.012	-0.018	-0.006
		(0.018)	(0.024)	(0.027)		(0.020)	(0.026)	(0.031)
Mother went to high school		0.031*	0.014	0.043		0.027	0.014	0.039
		(0.019)	(0.026)	(0.028)		(0.021)	(0.028)	(0.032)
Father went to high school		-0.032*	0.013	-0.072***		-0.024	-0.030	-0.018
		(0.017)	(0.024)	(0.025)		(0.019)	(0.026)	(0.029)
Father went to college		-0.006	-0.009	-0.012		-0.010	0.003	-0.017
		(0.020)	(0.026)	(0.032)		(0.022)	(0.028)	(0.037)
Locus of control		0.058	0.094	0.022		0.001	0.017	0.000
		(0.062)	(0.085)	(0.091)		(0.069)	(0.091)	(0.105)
Self-efficacy		-0.031	-0.057	-0.007		-0.056	-0.083	-0.037
		(0.047)	(0.064)	(0.070)		(0.053)	(0.069)	(0.081)
Spanish grade in 2022/23		0.003	0.002	0.006		0.000	0.003	-0.003
		(0.008)	(0.011)	(0.012)		(0.009)	(0.012)	(0.014)
Math grade in 2022/23		-0.010	-0.005	-0.018		0.008	0.013	0.000
		(0.009)	(0.012)	(0.014)		(0.010)	(0.013)	(0.016)
Constant	0.916***	1.075***	1.228***	0.881***	0.900***	1.239***	1.345***	1.181***
	(0.011)	(0.168)	(0.218)	(0.263)	(0.011)	(0.187)	(0.233)	(0.303)
R-squared	0.000	0.025	0.027	0.048	0.003	0.024	0.038	0.022
N	1333	991	493	498	1333	991	493	498

Notes: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Standard errors in parentheses.

Table 3: Treatment Effects on Math Assessments

	Midline		Endline	
	(1)	(2)	(3)	(4)
CAL Treatment (β_1)	0.2060	0.1673	0.3126**	0.2848**
	(0.1396)	(0.1205)	(0.1574)	(0.1387)
CAL × Tutoring (β_2)	-0.1034	-0.0663	-0.2395	-0.1943
	(0.1539)	(0.1261)	(0.1642)	(0.1442)
CAL + Tutoring ($\beta_1 + \beta_2$)	0.1026	0.1010	0.0731	0.0905
	(0.1318)	(0.1139)	(0.1446)	(0.1264)
N	1225	1225	1218	1218
Adjusted R^2	0.005	0.277	0.015	0.301
Covariates	No	Yes	No	Yes

Notes: Randomization inference standard errors in parentheses (1000 simulations). Covariates include students' gender, age, baseline math test scores, proxies for confidence and preferences towards math, self-reported grades for the previous year and parents' employment status. When available, we also control for the baseline level of the outcome variable. * p<0.1, ** p<0.05, *** p<0.01 based on RI p-values.

Table 4: Treatment Effects on Test Scores by Content Level

	Below Grade Level		At Grade Level	
	(1)	(2)	(3)	(4)
CAL Treatment (β_1)	0.2817**	0.2621**	0.2429*	0.2143*
	(0.1371)	(0.1266)	(0.1412)	(0.1206)
CAL × Tutoring (β_2)	-0.1630	-0.1269	-0.2617	-0.2199
	(0.1413)	(0.1334)	(0.1573)	(0.1323)
CAL + Tutoring ($\beta_1 + \beta_2$)	0.1187	0.1352	-0.0187	-0.0056
	(0.1268)	(0.1167)	(0.1299)	(0.1102)
N	1218	1218	1217	1217
Adjusted R^2	0.012	0.217	0.011	0.239
Covariates	No	Yes	No	Yes

Notes: Randomization inference standard errors in parentheses (1000 simulations). Covariates include students' gender, age, baseline math test scores, proxies for confidence and preferences towards math, self-reported grades for the previous year and parents' employment status. When available, we also control for the baseline level of the outcome variable. * p<0.1, ** p<0.05, *** p<0.01 based on RI p-values.

Table 5: Impact on Test Scores by Math Area

	Midline			Endline			
	Algebra	Numbers	Data & Prob	Algebra	Numbers	Data & Prob	Geometry
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
CAL Treatment (β_1)	0.195*	0.093	-0.011	0.214*	0.317**	0.028	0.133
	(0.113)	(0.115)	(0.064)	(0.121)	(0.143)	(0.087)	(0.115)
CAL × Tutoring (β_2)	-0.091	0.011	-0.062	-0.220	-0.106	0.056	-0.191
	(0.111)	(0.126)	(0.082)	(0.132)	(0.148)	(0.098)	(0.126)
CAL + Tutoring ($\beta_1 + \beta_2$)	0.104	0.104	-0.073	-0.006	0.210	0.084	-0.058
	(0.107)	(0.107)	(0.066)	(0.110)	(0.126)	(0.081)	(0.110)
N	1225	1225	1220	1217	1217	1218	1218
Adjusted R^2	0.189	0.187	0.055	0.239	0.179	0.044	0.080

Notes: All specifications include covariates. Randomization inference standard errors in parentheses (1000 simulations). Covariates include students' gender, age, baseline math test scores, proxies for confidence and preferences towards math, self-reported grades for the previous year and parents' employment status. When available, we also control for the baseline level of the outcome variable. * p<0.1, ** p<0.05, *** p<0.01 based on RI p-values.

Table 6: Treatment Effects on Test Scores by Knowledge Domain

	Midline			Endline		
	Knowing	Applying	Reasoning	Knowing	Applying	Reasoning
	(1)	(2)	(3)	(4)	(5)	(6)
CAL Treatment (β_1)	0.198	0.075	0.075	0.311***	0.164	0.211**
	(0.150)	(0.087)	(0.081)	(0.120)	(0.125)	(0.100)
CAL × Tutoring (β_2)	0.009	-0.064	-0.156*	-0.213	-0.097	-0.089
	(0.150)	(0.094)	(0.084)	(0.129)	(0.141)	(0.094)
CAL + Tutoring ($\beta_1 + \beta_2$)	0.207	0.010	-0.081	0.098	0.066	0.123
	(0.137)	(0.085)	(0.073)	(0.111)	(0.113)	(0.096)
N	1225	1225	1224	1218	1217	1218
Adjusted R^2	0.179	0.180	0.076	0.185	0.173	0.101

Notes: All specifications include covariates. Randomization inference standard errors in parentheses (1000 simulations). Covariates include students' gender, age, baseline math test scores, proxies for confidence and preferences towards math, self-reported grades for the previous year and parents' employment status. When available, we also control for the baseline level of the outcome variable. * p<0.1, ** p<0.05, *** p<0.01 based on RI p-values.

Table 7: Treatment Effects on Math Overconfidence

	Midline		Endline	
	(1)	(2)	(3)	(4)
CAL Treatment (β_1)	0.0181	0.0322	-0.0022	0.0083
	(0.0454)	(0.0447)	(0.0327)	(0.0345)
CAL × Tutoring (β_2)	0.0260	0.0290	0.0420	0.0502
	(0.0390)	(0.0414)	(0.0301)	(0.0294)
CAL + Tutoring ($\beta_1 + \beta_2$)	0.0441	0.0612	0.0398	0.0585*
	(0.0414)	(0.0418)	(0.0324)	(0.0338)
N	1205	1205	1188	1188
Control Mean	0.562	0.562	0.581	0.581
Adjusted R^2	-0.000	0.051	-0.000	0.055
Covariates	No	Yes	No	Yes

Notes: The outcome variable takes the value of 1 if the teacher switched classes after program assignment, and 0 otherwise. Randomization inference standard errors in parentheses (1000 simulations). Covariates include students' gender, age, baseline math test scores, proxies for confidence and preferences towards math, self-reported grades for the previous year and parents' employment status. When available, we also control for the baseline level of the outcome variable. * p<0.1, ** p<0.05, *** p<0.01 based on RI p-values.

Table 8: Dose-response of ALEKS Use

	Standardized Math Scores		
	IV estimates (1)	OLS VA (full sample) (2)	OLS VA (treatment group) (3)
CAL hours	0.0266***	0.0191***	0.0256***
	(0.0057)	(0.0032)	(0.0051)
Baseline math score (SD)	0.1431***	0.1456***	0.1160***
	(0.0100)	(0.0098)	(0.0145)
Observations	1218	1218	617
R-squared	0.3248	0.3296	0.3211
Difference-in-Sargan statistic for exogeneity (p-value)	0.1182		
Extrapolated estimates of 2.5-hour treatment	0.0665	0.0477	0.0641
Extrapolated estimates 25 hours (1500 minutes)	0.6649	0.4770	0.6409

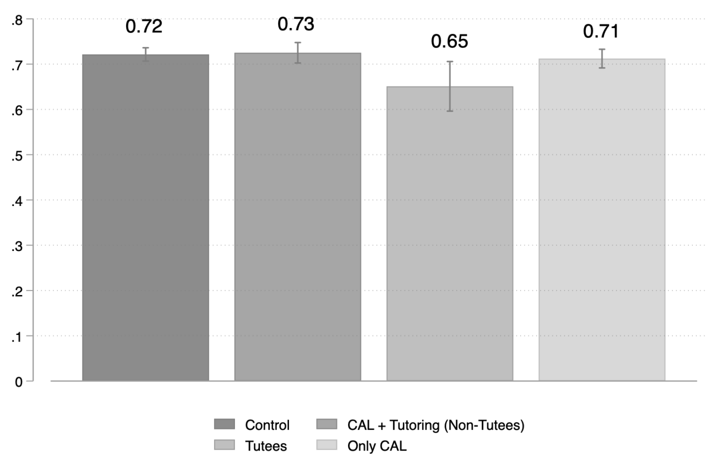
Notes: Robust standard errors in parentheses. * p<0.1, ** p<0.05, *** p<0.01. Treatment group students who were randomly-selected for the ALEKS platform offer but who did not take up the offer have been marked as having 0 sessions, as have all students in the control group. Covariates include students' gender, age, baseline math test scores, proxies for confidence and preferences towards math, self-reported grades for the previous year and parents' employment status. Column (1) instruments sessions in ALEKS with the randomized allocation. Column (2) presents OLS value-added models in the full sample. Column (3) presents OLS value-added models using only data on the lottery-winners. One hour is equivalent to 60 minutes of platform use.

Table A1: Lee Bounds - Single Regression Approach

	Midline Math Test		Endline Math Test	
	Lower Bound	Upper Bound	Lower Bound	Upper Bound
	(1)	(2)	(3)	(4)
CAL Treatment (β_1)	0.160	0.246*	0.233	0.379**
	(0.144)	(0.143)	(0.166)	(0.157)
CAL x Tutoring (β_2)	-0.175	-0.018	-0.252	-0.232
	(0.153)	(0.162)	(0.164)	(0.174)
CAL + Tutoring ($\beta_1 + \beta_2$)	0.071	0.142	0.001	0.127
	(0.136)	(0.135)	(0.151)	(0.148)
Observations (before trimming)	1225		1218	
Observations (after trimming)	1206		1199	

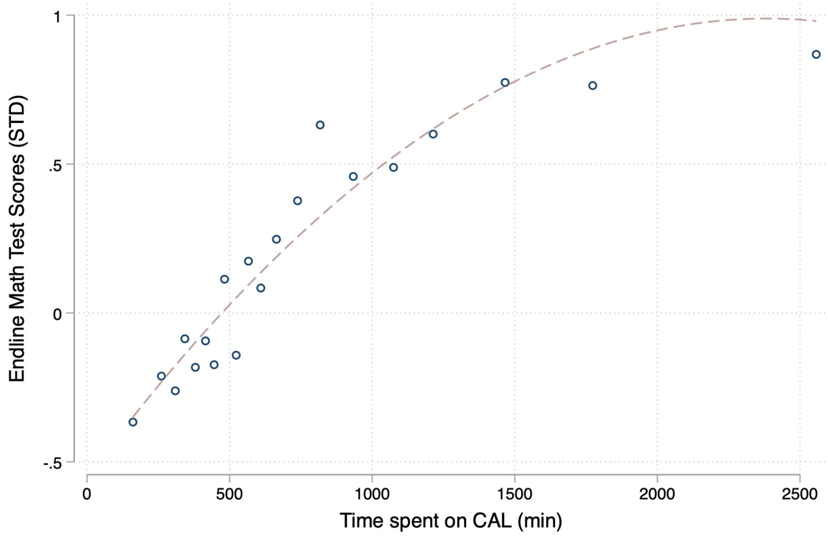
Notes: Lee (2009) bounds calculated using single regression approach with all three groups trimmed to minimum response rate (89.8%). Randomization inference standard errors in parentheses (1000 simulations). * p<0.1, ** p<0.05, *** p<0.01 based on RI p-values.

Figure A1: Self-Efficacy Index by Performance Groups



Notes: The CAL + Tutoring group excludes tutees.

Figure A2: Returns to Hours of CAL Use



Notes: This sample is limited to treated students (N = 664).

A Additional Analysis

A.1 Power Calculations

To estimate statistical power, we simulate the randomization procedure described in Section 1.3 one thousand times and assume a minimum detectable effect (MDE) of 0.3 standard deviations. Evans and Yuan (2022) find this to be in the 80th percentile of effect sizes of RCTs with learning or access outcomes in low- and middle-income countries. Although this would be above the median of 0.2 for small RCTs, both tutoring and computer-adaptive learning have been identified as two of the most effective interventions to improve learning outcomes. A recent review of tutoring interventions by Nickow et al. (2020) found an overall pooled effect size estimate of 0.37 standard deviations. To the best of our knowledge, there are no meta-studies looking at effects from computer-*adaptive* learning technologies such as the one used in this study.

For this exercise, I use 2019 administrative data on 9th grade students in the same sample of schools selected for this intervention. This data includes students' test scores on a national diagnostic assessment, their sex, age, and a socioeconomic indicator. We also simulate and include a baseline test score variable. These simulations give us a statistical power of approximately 80 percent, which is the standard for randomized controlled trials in the social sciences. We aim to maximize this number by collecting prognostic variables at baseline, including tests scores on a short math assessment and other student characteristics. In Table A2, we present the results of the simulations using alternative MDEs (0.25, 0.3, 0.35).

MDE	Power
0.25	0.674
0.30	0.799
0.35	0.893

Table A2: Power Simulations

B Additional Results

This appendix presents detailed results for secondary outcomes summarized in Section 5.4.

B.1 Intrinsic Motivation, Absenteeism, and Dropout

Beyond academic performance, improvements in learning could also positively influence students' retention in school by increasing their motivation and interest in the subject, as well as their chances of obtaining a passing grade. In Tables B1, B2 and B3, we present impacts on intrinsic motivation, preferences towards math, absenteeism, and student dropout. Student dropout is a significant concern in our context, especially among the targeted age group. In the Dominican Republic, over 40 percent of students drop out between the 8th and 12th grades (IDEICE, 2017). In the control group, 8 percent of students dropped out of school during the intervention period (see Table B3). However, we find no evidence that the intervention impacted students' intrinsic motivation, how much they like math (self-reported 1 to 4 scale), school absenteeism, or dropout rates by endline. The null effect on dropout is consistent with the possibility that, in this context, students face binding non-academic constraints—such as economic pressures or family obligations—that learning gains alone cannot address. The lack of findings on intrinsic motivation align with Araya et al. (2019), who also observed no effect on this outcome in a study using a game-based learning platform in Chile.

B.2 Math Self-Concept

As students spend more time on the software and master new topics, they receive new information on their abilities and may update their beliefs accordingly. In Table B4, we present effects on students' math self-concept, defined as students' perceptions of their own ability in mathematics compared to others. As in Araya et al. (2019), despite the large increases observed in math performance, we do not find statistically significant differences across experimental groups in a math self-concept index or on individual measures used to build the index. However, students in the CAL+Tutoring group were 9.3 percentage points less likely to agree with the statement "Math is harder for me than for many of my classmates" than students in the control group or in the CAL-only group. While these results are not conclusive, they may be suggestive of shifts in students' self-perception and confidence in their math abilities. Despite performing worse on math tests, the tutoring group seems to feel equally confident about their performance and are more likely to have mismatched expectations about their outcomes in the test, when compared to those whose classrooms were not assigned to the tutoring component.

B.3 Self-Efficacy, Grit, and Perseverance

By offering personalized learning pathways and continuous feedback on individual progress and milestones, the platform allows students to understand and learn from their mistakes privately, without the stigma of public correction. As students master new topics and receive real-time feedback, this process may enhance their sense of competence and self-efficacy. In this subsection, we examine four related outcomes: self-efficacy, grit, perseverance, and locus of control. As shown in Table B5, neither the CAL intervention nor the tutoring intervention had a significant impact on a self-efficacy index.[21] This suggests that the interventions did not notably alter students' beliefs in their ability to achieve academic success.

Similarly, we do not find impacts on a measure of grit (see Table B6), which assesses students' perseverance and passion for long-term goals (Duckworth et al., 2007). Following Alan et al. (2019), we implement an alternative measure of perseverance, based on a real-effort exercise. Table B7 shows that treated students are 8.7 percentage points more likely to choose to undertake an easier task against a more challenging alternative. We do not observe differences in their likelihood of giving up after failure for either group. This suggests that the adaptive nature of the platform, which presents exercises based on students' current ability, may have negatively impacted their willingness to take on more challenging exercises.

B.4 Student Wellbeing

To assess changes on students' wellbeing, we use two proxies: the Children's Depression Screener (ChilD-S) and a a short questionnaire on math anxiety. For the first one, we build an index based on students' answers to 8 items about their mental state (e.g., "I feel happy", "I feel lonely", "It is hard for me to concentrate"). For the second one, we use two questions on students' worry about their math grades and their feelings before math tests to build a math anxiety index. Table B9 shows that the intervention had no impact on the first measure of wellbeing. While we do not find impacts on a math anxiety index, we find important differences when looking at individual measures of math anxiety. As seen in Table B10, students in the CAL-only group were 6 percentage points more likely to get nervous before math tests than students in the control and CAL+Tutoring groups. This difference is statistically significant at the 5 percent level. While math anxiety is typically linked to lower performance (Dowker et al., 2016), this result may be positive if it signals higher levels of awareness or concern about their math performance in this group.

[21] This index is built using students' answers to the following items: 1) I can always solve difficult problems if I try hard enough, 2) I can solve most problems if I invest the necessary effort, 3) I can stay calm when I face difficulties because I can trust my coping skills, 4) When faced with a problem, I can usually find several solutions, and 5) If I'm in trouble I can find a solution.

B.5 Teacher Practices

The CAL platform may free up teachers' time by automating tasks such as developing and grading homework and tests. Additionally, the presence of tutors in CAL+Tutoring classrooms may affect teacher behavior through increased accountability or organizational demands. Table B11 presents effects on indices of teacher time mismanagement and classroom engagement, constructed from student reports.

We find that classrooms assigned to CAL+Tutoring showed significantly lower teacher time mismanagement relative to the control group (0.23 SD without controls, $p<0.05$; 0.18 SD with controls, $p<0.10$). The time mismanagement index captures whether teachers were late to class, absent from school, started or finished class early or late, or left school early. The significant reduction in this index suggests that teachers in CAL+Tutoring classrooms were more punctual and had fewer absences. This pattern may reflect increased accountability from the presence of tutors, or the greater organizational structure required to coordinate the combined intervention.

For teacher engagement, CAL-only classrooms showed marginally lower scores relative to control (0.18 SD, $p<0.10$ without controls), though this effect is not robust to the inclusion of covariates. This pattern may reflect that the platform substitutes for some direct teacher instruction during CAL sessions. We do not observe significant differences in teacher engagement for the CAL+Tutoring group.

Platform data provide additional context: while teachers spent an average of 33 minutes per week actively using the platform, they seldom leveraged it to automate routine tasks. Over the course of the academic year, teachers assigned no more than three activities through the CAL platform, suggesting limited adoption of the platform's labor-saving features.

B.6 Student Time Use

One potential concern with technology-based interventions is that they might displace other productive activities or increase screen time outside of school. To examine whether the intervention affected how students allocate their time, we collected data on hours spent in eight categories: in-person classes, unpaid work at home, unpaid work outside the home, paid work, studying outside of school, meeting friends, entertainment, and sports.

Table B12 presents treatment effects on student time use. We find no statistically significant effects on any of these categories for either the CAL-only or CAL+Tutoring groups. The null effects suggest that the intervention operated primarily through the substitution of regular math instruction during school hours rather than through changes in how students allocated time outside of CAL sessions.

Table B1: Treatment Effects on Student Motivation and Attitudes

	Intrinsic Motivation (1)	Math Enjoyment (2)	Math Confidence (3)	Math Importance (4)
CAL Treatment (β_1)	-0.0257	0.0357	-0.0024	-0.1334**
	(0.0807)	(0.0678)	(0.0864)	(0.0626)
CAL × Tutoring (β_2)	0.1020	0.0707	-0.0090	0.0927
	(0.0899)	(0.0811)	(0.0935)	(0.0835)
CAL + Tutoring ($\beta_1 + \beta_2$)	0.0763	0.1065	-0.0113	-0.0407
	(0.0746)	(0.0671)	(0.0745)	(0.0628)
N	1194	1191	1179	1181
Control Mean	-0.000	2.733	2.327	2.744
Adjusted R^2	0.101	0.084	0.037	0.064

Notes: Randomization inference standard errors in parentheses (1000 simulations). All regressions include baseline covariates: student gender, age, baseline math score, parental employment status, and student confidence in math. * p<0.1, ** p<0.05, *** p<0.01 based on RI p-values.

Table B2: Treatment Effects on Math Preferences

	Midline		Endline	
	(1)	(2)	(3)	(4)
CAL Treatment (β_1)	-0.0511	0.0053	-0.0212	0.0203
	(0.0847)	(0.0747)	(0.0914)	(0.0870)
CAL × Tutoring (β_2)	-0.0299	-0.0378	0.0228	0.0235
	(0.1043)	(0.0851)	(0.1138)	(0.1018)
CAL + Tutoring ($\beta_1 + \beta_2$)	-0.0810	-0.0325	0.0016	0.0438
	(0.0770)	(0.0677)	(0.0828)	(0.0789)
N	1218	1218	1210	1210
Control Mean	2.793	2.793	2.676	2.676
Adjusted R^2	-0.000	0.322	-0.002	0.235
Covariates	No	Yes	No	Yes

Notes: Randomization inference standard errors in parentheses (1000 simulations). Covariates include students' gender, age, baseline math test scores, proxies for confidence and preferences towards math, self-reported grades for the previous year and parents' employment status. When available, we also control for the baseline level of the outcome variable. * p<0.1, ** p<0.05, *** p<0.01 based on RI p-values.

Table B3: Treatment Effects on Absence and School Dropout

	Absence		Dropout	
	(1)	(2)	(3)	(4)
CAL Treatment (β_1)	-1.4203	-0.6303	-0.0234	-0.0142
	(2.3424)	(1.9138)	(0.0231)	(0.0186)
CAL $\times$ Tutoring (β_2)	1.9477	1.1173	-0.0059	-0.0133
	(2.8614)	(2.3589)	(0.0291)	(0.0233)
CAL + Tutoring ($\beta_1 + \beta_2$)	0.5275	0.4870	-0.0293	-0.0274
	(2.0838)	(1.7021)	(0.0218)	(0.0173)
N	1333	1333	1333	1333
Control Mean	10.614	10.614	0.077	0.077
Adjusted R^2	0.002	0.138	0.001	0.056
Covariates	No	Yes	No	Yes

Notes: Randomization inference standard errors in parentheses (1000 simulations). Attendance measures are specific to math classes. Covariates include students' gender, age, baseline math test scores, proxies for confidence and preferences towards math, self-reported grades for the previous year and parents' employment status. When available, we also control for the baseline level of the outcome variable. * p<0.1, ** p<0.05, *** p<0.01 based on RI p-values.

Table B4: Treatment Effects on Math Self-Concept

	Math Self-Concept Index (1)	I usually do well in math (2)	Math is harder for me than for many of my classmates (3)
CAL Treatment (β_1)	-0.0076	-0.0053	0.0324
	(0.0393)	(0.0870)	(0.0712)
CAL $\times$ Tutoring (β_2)	0.0109	-0.0029	-0.0928
	(0.0411)	(0.1071)	(0.0749)
CAL + Tutoring ($\beta_1 + \beta_2$)	0.0034	-0.0081	-0.0604
	(0.0342)	(0.0859)	(0.0604)
N	1184	1182	1175
Control Mean	0.694	2.663	2.519
Adjusted R^2	0.160	0.164	0.063

Notes: Randomization inference standard errors in parentheses (1000 simulations). Covariates include students' gender, age, baseline math test scores, proxies for confidence and preferences towards math, self-reported grades for the previous year and parents' employment status. When available, we also control for the baseline level of the outcome variable. * p<0.1, ** p<0.05, *** p<0.01 based on RI p-values.

Table B5: Treatment Effects on Self-Efficacy Index

	Midline		Endline	
	(1)	(2)	(3)	(4)
CAL Treatment (β_1)	-0.0162	-0.0179	-0.0092	-0.0120
	(0.0141)	(0.0119)	(0.0120)	(0.0111)
CAL $\times$ Tutoring (β_2)	0.0171	0.0200*	-0.0004	0.0057
	(0.0156)	(0.0120)	(0.0160)	(0.0155)
CAL + Tutoring ($\beta_1 + \beta_2$)	0.0009	0.0021	-0.0096	-0.0063
	(0.0130)	(0.0116)	(0.0131)	(0.0132)
N	1217	1217	1194	1194
Control Mean	0.767	0.767	0.721	0.721
Adjusted R^2	0.000	0.248	-0.001	0.129
Covariates	No	Yes	No	Yes

Notes: Randomization inference standard errors in parentheses (1000 simulations). Covariates include students' gender, age, baseline math test scores, proxies for confidence and preferences towards math, self-reported grades for the previous year and parents' employment status. When available, we also control for the baseline level of the outcome variable. * $p<0.1$, ** $p<0.05$, *** $p<0.01$ based on RI p-values.

Table B6: Treatment Effects on Grit Index

	Endline	
	(1)	(2)
CAL Treatment (β_1)	0.0110	0.0055
	(0.0137)	(0.0126)
CAL $\times$ Tutoring (β_2)	0.0009	0.0007
	(0.0162)	(0.0144)
CAL + Tutoring ($\beta_1 + \beta_2$)	0.0119	0.0062
	(0.0128)	(0.0118)
N	1333	1333
Control Mean	0.469	0.469
Adjusted R^2	-0.001	0.033
Covariates	No	Yes

Notes: The outcome is a standardized variable based on students' answers to the following items: 1) The schoolwork I like the most is the one that makes me think, 2) If I think I will lose in a game, I do not want to continue playing, 3) If I set a goal and see that it's harder than I though I lose interest, and 4) I work hard on tasks. Answer options ranged from 1 (Doesn't seem like me at all) to 5 (Very much like me). Randomization inference standard errors in parentheses (1000 simulations). Covariates include students' gender, age, baseline math test scores, proxies for confidence and preferences towards math, self-reported grades for the previous year and parents' employment status. When available, we also control for the baseline level of the outcome variable. * p<0.1, ** p<0.05, *** p<0.01 based on RI p-values.

Table B7: Treatment Effects on Logic Task

	Easy	Difficult	Give Up
	(1)	(2)	(3)
CAL Treatment (β_1)	0.0868*	-0.0939**	0.0072
	(0.0474)	(0.0428)	(0.0270)
CAL $\times$ Tutoring (β_2)	-0.0367	0.0231	0.0136
	(0.0550)	(0.0554)	(0.0312)
CAL + Tutoring ($\beta_1 + \beta_2$)	0.0501	-0.0708*	0.0207
	(0.0461)	(0.0407)	(0.0244)
N	1211	1211	1211
Control Mean	0.435	0.380	0.186
Adjusted R^2	0.067	0.072	-0.001

Notes: Randomization inference standard errors in parentheses (1000 simulations). Covariates include students' gender, age, baseline math test scores, proxies for confidence and preferences towards math, self-reported grades for the previous year and parents' employment status. When available, we also control for the baseline level of the outcome variable. * p<0.1, ** p<0.05, *** p<0.01 based on RI p-values.

Table B8: Treatment Effects on Locus of Control

	Endline	
	(1)	(2)
CAL Treatment (β_1)	0.0104	0.0078
	(0.0087)	(0.0081)
CAL $\times$ Tutoring (β_2)	-0.0245*	-0.0216*
	(0.0110)	(0.0108)
CAL + Tutoring ($\beta_1 + \beta_2$)	-0.0142	-0.0138*
	(0.0081)	(0.0079)
N	1196	1196
Control Mean	0.667	0.667
Adjusted R^2	0.003	0.073
Covariates	No	Yes

Notes: Randomization inference standard errors in parentheses (1000 simulations). Covariates include students' gender, age, baseline math test scores, proxies for confidence and preferences towards math, self-reported grades for the previous year and parents' employment status. When available, we also control for the baseline level of the outcome variable. * p<0.1, ** p<0.05, *** p<0.01 based on RI p-values.

Table B9: Treatment Effects on Student Depression

	Depression Index	
	(1)	(2)
CAL Treatment (β_1)	0.0120	0.0092
	(0.0109)	(.)
CAL × Tutoring (β_2)	-0.0026	-0.0032
	(0.0112)	(.)
CAL + Tutoring ($\beta_1 + \beta_2$)	0.0093	0.0060
	(0.0105)	(.)
N	1205	1205
Control Mean	0.612	0.612
Adjusted R^2	-0.000	0.082
Covariates	No	Yes

Notes: The outcome variable is a standardized index based on students' answers to the following items: 1) I am happy, 2) I worry a lot, 3) I feel sad, 4) I get upset quickly, 5) I often think I did something wrong, 6) It's hard for me to concentrate, 7) I feel lonely, and 8) I am not happy. Answers range from 1 (Strongly disagree) to 4 (Strongly agree). Randomization inference standard errors in parentheses (1000 simulations). Covariates include students' gender, age, baseline math test scores, proxies for confidence and preferences towards math, self-reported grades for the previous year, and parents' employment status. When available, we also control for the baseline level of the outcome variable. * p<0.1, ** p<0.05, *** p<0.01 based on RI p-values.

Table B10: Treatment Effects on Math Anxiety

	I'm worried of having bad grades in math	I get nervous before math tests	Math Anxiety Index
	(1)	(2)	(3)
CAL Treatment (β_1)	0.0330	0.0599	0.1350
	(0.0333)	(0.0371)	(0.0939)
CAL × Tutoring (β_2)	-0.0157	-0.0606	-0.1226
	(0.0348)	(0.0375)	(0.0938)
CAL + Tutoring ($\beta_1 + \beta_2$)	0.0173	-0.0007	0.0124
	(0.0291)	(0.0372)	(0.0888)
N	1199	1183	1182
Control Mean	0.774	0.713	0.000
Adjusted R^2	0.028	0.036	0.036

Notes: Randomization inference standard errors in parentheses (100 simulations). Covariates include students' gender, age, baseline math test scores, proxies for confidence and preferences towards math, self-reported grades for the previous year and parents' employment status. When available, we also control for the baseline level of the outcome variable. * p<0.1, ** p<0.05, *** p<0.01 based on RI p-values.

Table B11: Treatment Effects on Teacher Behavior Indices

	Time Mismanagement		Teacher Engagement	
	(1)	(2)	(3)	(4)
CAL Treatment (β_1)	-0.0816	-0.0304	-0.1808*	-0.1437
	(0.1252)	(0.1163)	(0.1038)	(0.0882)
CAL × Tutoring (β_2)	-0.1455	-0.1458	0.0442	0.0407
	(0.1164)	(0.1071)	(0.1109)	(0.1054)
CAL + Tutoring ($\beta_1 + \beta_2$)	-0.2271**	-0.1763*	-0.1366	-0.1030
	(0.1119)	(0.1076)	(0.0929)	(0.0769)
N	1148	1148	1151	1151
Adjusted R^2	0.007	0.117	0.005	0.116
Covariates	No	Yes	No	Yes

Notes: Randomization inference standard errors in parentheses (1000 simulations). Covariates include students' gender, age, baseline math test scores, proxies for confidence and preferences towards math, self-reported grades for the previous year and parents' employment status. When available, we also control for the baseline level of the outcome variable. * p<0.1, ** p<0.05, *** p<0.01 based on RI p-values.

Table B12: Treatment Effects on Student Time Use

	In person classes	Unpaid work (at home)	Unpaid work (out of home)	Paid work (out of school)	Study	Meeting friends	Entertain-ment	Sports
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
CAL Treatment (β_1)	0.0286	-0.2489	-0.0004	0.0237	0.0515	0.0020	0.0435	0.0321
	(0.2394)	(0.2189)	(0.2208)	(0.2470)	(0.1465)	(0.1281)	(0.0868)	(0.0977)
CAL × Tutoring (β_2)	-0.2366	0.0183	-0.1347	-0.2455	0.0644	-0.1247	-0.0075	0.0205
	(0.2869)	(0.2369)	(0.2317)	(0.2819)	(0.1512)	(0.1859)	(0.1050)	(0.1217)
CAL + Tutoring ($\beta_1 + \beta_2$)	-0.2080	-0.2307	-0.1352	-0.2218	0.1158	-0.1226	0.0361	0.0525
	(0.2363)	(0.2007)	(0.1974)	(0.2322)	(0.1310)	(0.1165)	(0.0824)	(0.0919)
Control Mean	5.99	3.79	2.26	2.74	2.03	2.25	3.33	2.09
Adjusted R^2	0.06	0.05	0.11	0.11	0.01	0.04	0.01	0.13
Observations	1182	1168	1156	1152	1177	1169	1169	1166

Notes: * p<0.01, ** p<0.05, *** p<0.1. Covariates include student's gender, age, baseline math test scores, proxies for confidence and preferences towards math, self-reported grades for the previous year and parents' employment status. Randomization inference standard errors in parentheses (1000 simulations).